```
  LOC       OBJECT CODE      ADDR1     ADDR2    STMT

                                                   2 ***********************************************************************
                                                   3 *
                                                   4 *   Zvector E7 instruction tests for VRR-c encoded:
                                                   5 *
                                                   6 *        E7F0 VAVGL  - VECTOR AVERAGE LOGICAL
                                                   7 *        E7F2 VAVG   - VECTOR AVERAGE
                                                   8 *        E7FC VMNL   - VECTOR MINIMUM LOGICAL
                                                   9 *        E7FD VMXL   - VECTOR MAXIMUM LOGICAL
                                                  10 *        E7FE VMN    - VECTOR MINIMUM
                                                  11 *        E7FF VMX    - VECTOR MAXIMUM
                                                  12 *
                                                  13 *        James Wekel July 2024
                                                  14 *                    July 2025 - Vector-enhancements facility 3 update
                                                  15 ***********************************************************************

                                                  17 ***********************************************************************
                                                  18 *
                                                  19 *        basic instruction tests
                                                  20 *
                                                  21 ***********************************************************************
                                                  22 *  This program tests proper functioning of the z/arch E7 VRR-c vector
                                                  23 *  average, minimum and maximum instructions.
                                                  24 *  Exceptions are not tested.
                                                  25 *
                                                  26 *  PLEASE NOTE that the tests are very SIMPLE TESTS designed to catch
                                                  27 *  obvious coding errors.  None of the tests are thorough.  They are
                                                  28 *  NOT designed to test all aspects of any of the instructions.
                                                  29 *
                                                  30 ***********************************************************************
                                                  31 *
                                                  32 *   *Testcase zvector-e7-01-MinMaxAvg: VECTOR E7 VRR-c instructions
                                                  33 *   *
                                                  34 *   *   Zvector E7 instruction tests for VRR-c encoded:
                                                  35 *   *
                                                  36 *   *   E7F0 VAVGL  - VECTOR AVERAGE LOGICAL
                                                  37 *   *   E7F2 VAVG   - VECTOR AVERAGE
                                                  38 *   *   E7FC VMNL   - VECTOR MINIMUM LOGICAL
                                                  39 *   *   E7FD VMXL   - VECTOR MAXIMUM LOGICAL
                                                  40 *   *   E7FE VMN    - VECTOR MINIMUM
                                                  41 *   *   E7FF VMX    - VECTOR MAXIMUM
                                                  42 *   *
                                                  43 *   *   # ------------------------------------------------------------
                                                  44 *   *   #  This tests only the basic function of the instruction.
                                                  45 *   *   #  Exceptions are NOT tested.
                                                  46 *   *   # ------------------------------------------------------------
                                                  47 *   *
                                                  48 *   mainsize    2
                                                  49 *   numcpu      1
                                                  50 *   sysclear
                                                  51 *   archlvl     z/Arch
                                                  52 *
                                                  53 *   loadcore    "$(testpath)/zvector-e7-01-MinMaxAvg.core" 0x0
                                                  54 *
                                                  55 *   diag8cmd    enable    # (needed for messages to Hercules console)
                                                  56 *   runtest     2
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                    57 *   diag8cmd    disable   # (reset back to default)
                                                    58 *
                                                    59 *   *Done
                                                    60 *
                                                    61 ***********************************************************************
```

```
  LOC      OBJECT CODE     ADDR1    ADDR2    STMT

                                               63 *********************************************************************
                                               64 *        FCHECK Macro - Is a Facility Bit set?
                                               65 *
                                               66 *        If the facility bit is NOT set, an message is issued and
                                               67 *        the test is skipped.
                                               68 *
                                               69 *        Fcheck uses R0, R1 and R2
                                               70 *
                                               71 * eg.    FCHECK 134,'vector-packed-decimal'
                                               72 *********************************************************************
                                               73          MACRO
                                               74          FCHECK &BITNO,&NOTSETMSG
                                               75 .*                         &BITNO : facility bit number to check
                                               76 .*                         &NOTSETMSG : 'facility name'
                                               77          LCLA  &FBBYTE           Facility bit in Byte
                                               78          LCLA  &FBBIT            Facility bit within Byte
                                               79
                                               80          LCLA  &L(8)
                                               81 &L(1)    SetA  128,64,32,16,8,4,2,1  bit positions within byte
                                               82
                                               83 &FBBYTE  SETA  &BITNO/8
                                               84 &FBBIT   SETA  &L((&BITNO-(&FBBYTE*8))+1)
                                               85 .*       MNOTE 0,'checking Bit=&BITNO: FBBYTE=&FBBYTE, FBBIT=&FBBIT'
                                               86
                                               87          B     X&SYSNDX
                                               88 *                                      Fcheck data area
                                               89 *                                      skip messgae
                                               90 SKT&SYSNDX DC  C'    Skipping tests: '
                                               91          DC    C&NOTSETMSG
                                               92          DC    C' (bit &BITNO) is not installed.'
                                               93 SKL&SYSNDX EQU *-SKT&SYSNDX
                                               94 *                                      facility bits
                                               95          DS    FD                      gap
                                               96 FB&SYSNDX DS   4FD
                                               97          DS    FD                      gap
                                               98 *
                                               99 X&SYSNDX EQU *
                                              100          LA    R0,((X&SYSNDX-FB&SYSNDX)/8)-1
                                              101          STFLE FB&SYSNDX               get facility bits
                                              102
                                              103          XGR   R0,R0
                                              104          IC    R0,FB&SYSNDX+&FBBYTE    get fbit byte
                                              105          N     R0,=F'&FBBIT'           is bit set?
                                              106          BNZ   XC&SYSNDX
                                              107 *
                                              108 * facility bit not set, issue message and exit
                                              109 *
                                              110          LA    R0,SKL&SYSNDX           message length
                                              111          LA    R1,SKT&SYSNDX           message address
                                              112          BAL   R2,MSG
                                              113
                                              114          B     EOJ
                                              115 XC&SYSNDX EQU *
                                              116          MEND
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   118 ***********************************************************************
                                                   119 *        Low core PSWs
                                                   120 ***********************************************************************
                              00000000  00003253   121 ZVE7TST  START 0
00000000                      00000000             122          USING ZVE7TST,R0            Low core addressability
                                                   123
                              00000140  00000000   124 SVOLDPSW EQU   ZVE7TST+X'140'        z/Arch Supervisor call old PSW

00000000                      00000000  000001A0   126          ORG   ZVE7TST+X'1A0'        z/Architecure RESTART PSW
000001A0  00000001 80000000                        127          DC    X'0000000180000000'
000001A8  00000000 00000200                        128          DC    AD(BEGIN)

000001B0                      000001B0  000001D0   130          ORG   ZVE7TST+X'1D0'        z/Architecure PROGRAM CHECK PSW
000001D0  00020001 80000000                        131          DC    X'0002000180000000'
000001D8  00000000 0000DEAD                        132          DC    AD(X'DEAD')

000001E0                      000001E0  00000200   134          ORG   ZVE7TST+X'200'        Start of actual test program...

                                                   136 ***********************************************************************
                                                   137 *               The actual "ZVE7TST" program itself...
                                                   138 ***********************************************************************
                                                   139 *
                                                   140 *  Architecture Mode: z/Arch
                                                   141 *  Register Usage:
                                                   142 *
                                                   143 *   R0       (work)
                                                   144 *   R1-4     (work)
                                                   145 *   R5       Testing control table - current test base
                                                   146 *   R6-R7    (work)
                                                   147 *   R8       First base register
                                                   148 *   R9       Second base register
                                                   149 *   R10      Third base register
                                                   150 *   R11      E7TEST call return
                                                   151 *   R12      E7TESTS register
                                                   152 *   R13      (work)
                                                   153 *   R14      Subroutine call
                                                   154 *   R15      Secondary Subroutine call or work
                                                   155 *
                                                   156 ***********************************************************************

00000200                      00000200             158          USING  BEGIN,R8        FIRST Base Register
00000200                      00001200             159          USING  BEGIN+4096,R9   SECOND Base Register
00000200                      00002200             160          USING  BEGIN+8192,R10  THIRD Base Register

00000200  0580                                     162 BEGIN    BALR  R8,0             Initalize FIRST base register
00000202  0680                                     163          BCTR  R8,0             Initalize FIRST base register
00000204  0680                                     164          BCTR  R8,0             Initalize FIRST base register

00000206  4190 8800                     00000800   166          LA    R9,2048(,R8)     Initalize SECOND base register
0000020A  4190 9800                     00000800   167          LA    R9,2048(,R9)     Initalize SECOND base register
                                                   168
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

0000020E  41A0 9800                    00000800   169          LA    R10,2048(,R9)    Initalize THIRD base register
00000212  41A0 A800                    00000800   170          LA    R10,2048(,R10)   Initalize THIRD base register
                                                  171
00000216  B600 833C                    0000053C   172          STCTL R0,R0,CTLR0      Store CR0 to enable AFP
0000021A  9604 833D                    0000053D   173          OI    CTLR0+1,X'04'    Turn on AFP bit
0000021E  9602 833D                    0000053D   174          OI    CTLR0+1,X'02'    Turn on Vector bit
00000222  B700 833C                    0000053C   175          LCTL  R0,R0,CTLR0      Reload updated CR0
                                                  176
                                                  177 ***********************************************************************
                                                  178 * Is Vector packed-decimal facility installed  (bit 134)
                                                  179 ***********************************************************************
                                                  180
                                                  181          FCHECK 129,'z/Architecture vector facility'
00000226  47F0 80A8                    000002A8   182+         B     X0001
                                                  183+*                                      Fcheck data area
                                                  184+*                                      skip messgae
0000022A  40404040 E2928997                       185+SKT0001  DC    C'    Skipping tests: '
0000023E  A961C199 838889A3                       186+         DC    C'z/Architecture vector facility'
0000025C  404D8289 A340F1F2                       187+         DC    C' (bit 129) is not installed.'
                             0000004E  00000001   188+SKL0001  EQU   *-SKT0001
                                                  189+*                                      facility bits
00000278  00000000 00000000                       190+         DS    FD                      gap
00000280  00000000 00000000                       191+FB0001   DS    4FD
000002A0  00000000 00000000                       192+         DS    FD                      gap
                                                  193+*
                             000002A8  00000001   194+X0001    EQU   *
000002A8  4100 0004                    00000004   195+         LA    R0,((X0001-FB0001)/8)-1
000002AC  B2B0 8080                    00000280   196+         STFLE FB0001                  get facility bits
000002B0  B982 0000                               197+         XGR   R0,R0
000002B4  4300 8090                    00000290   198+         IC    R0,FB0001+16            get fbit byte
000002B8  5400 8344                    00000544   199+         N     R0,=F'64'               is bit set?
000002BC  4770 80D0                    000002D0   200+         BNZ   XC0001
                                                  201+*
                                                  202+* facility bit not set, issue message and exit
                                                  203+*
000002C0  4100 004E                    0000004E   204+         LA    R0,SKL0001              message length
000002C4  4110 802A                    0000022A   205+         LA    R1,SKT0001              message address
000002C8  4520 8258                    00000458   206+         BAL   R2,MSG
000002CC  47F0 8320                    00000520   207+         B     EOJ
                             000002D0  00000001   208+XC0001   EQU   *
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   210 **********************************************************************
                                                   211 * Is z/Architecture vector facility installed  (bit 198)
                                                   212 **********************************************************************
                                                   213
                                                   214          FCHECK 198,'Vector-enhancements facility 3'
000002D0  47F0 8158                       00000358 215+         B     X0002
                                                   216+*                                        Fcheck data area
                                                   217+*                                        skip messgae
000002D4  40404040 E2928997                        218+SKT0002  DC    C'    Skipping tests: '
000002E8  E58583A3 96996085                        219+         DC    C'Vector-enhancements facility 3'
00000306  404D8289 A340F1F9                        220+         DC    C' (bit 198) is not installed.'
                             0000004E  00000001    221+SKL0002  EQU   *-SKT0002
                                                   222+*                                        facility bits
00000328  00000000 00000000                        223+         DS    FD                        gap
00000330  00000000 00000000                        224+FB0002   DS    4FD
00000350  00000000 00000000                        225+         DS    FD                        gap
                                                   226+*
                             00000358  00000001    227+X0002    EQU   *
00000358  4100 0004                       00000004 228+         LA    R0,((X0002-FB0002)/8)-1
0000035C  B2B0 8130                       00000330 229+         STFLE FB0002                    get facility bits
00000360  B982 0000                                230+         XGR   R0,R0
00000364  4300 8148                       00000348 231+         IC    R0,FB0002+24              get fbit byte
00000368  5400 8348                       00000548 232+         N     R0,=F'2'                  is bit set?
0000036C  4770 8180                       00000380 233+         BNZ   XC0002
                                                   234+*
                                                   235+* facility bit not set, issue message and exit
                                                   236+*
00000370  4100 004E                       0000004E 237+         LA    R0,SKL0002                message length
00000374  4110 80D4                       000002D4 238+         LA    R1,SKT0002                message address
00000378  4520 8258                       00000458 239+         BAL   R2,MSG
0000037C  47F0 8320                       00000520 240+         B     EOJ
                             00000380  00000001    241+XC0002   EQU   *
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   243 ***********************************************************************
                                                   244 *              Do tests in the E7TESTS table
                                                   245 ***********************************************************************
                                                   246
00000380  58C0 834C                     0000054C   247          L     R12,=A(E7TESTS)       get table of test addresses
                                                   248
                              00000384  00000001   249 NEXTE6   EQU   *
00000384  5850 C000                     00000000   250          L     R5,0(0,R12)       get test address
00000388  1255                                     251          LTR   R5,R5                have a test?
0000038A  4780 81CE                     000003CE   252          BZ    ENDTEST                 done?
                                                   253
0000038E                      00000000             254          USING E7TEST,R5
                                                   255
0000038E  4800 5004                     00000004   256          LH    R0,TNUM           save current test number
00000392  5000 8E04                     00001004   257          ST    R0,TESTING        for easy reference
                                                   258
00000396  E710 8E94 0006                00001094   259          VL    V1,V1FUDGE
0000039C  58B0 5000                     00000000   260          L     R11,TSUB          get address of test routine
000003A0  05BB                                     261          BALR  R11,R11           do test
                                                   262
000003A2  E310 501C 0014                0000001C   263          LGF   R1,READDR         get address of expected result
000003A8  D50F 5028 1000      00000028  00000000   264          CLC   V1OUTPUT,0(R1)    valid?
000003AE  4770 81BA                     000003BA   265          BNE   FAILMSG              no, issue failed message
                                                   266
000003B2  41C0 C004                     00000004   267          LA    R12,4(0,R12)      next test address
000003B6  47F0 8184                     00000384   268          B     NEXTE6
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   270 ***********************************************************************
                                                   271 * result not as expected:
                                                   272 *        issue message with test number, instruction under test
                                                   273 *              and instruction m4
                                                   274 ***********************************************************************
                               000003BA  00000001  275 FAILMSG  EQU   *
000003BA  45F0 81DC                      000003DC  276          BAL   R15,RPTERROR

                                                   278 ***********************************************************************
                                                   279 * continue after a failed test
                                                   280 ***********************************************************************
                               000003BE  00000001  281 FAILCONT EQU   *
000003BE  5800 8350                      00000550  282          L     R0,=F'1'          set failed test indicator
000003C2  5000 8E00                      00001000  283          ST    R0,FAILED
                                                   284
000003C6  41C0 C004                      00000004  285          LA    R12,4(0,R12)      next test address
000003CA  47F0 8184                      00000384  286          B     NEXTE6

                                                   288 ***********************************************************************
                                                   289 * end of testing; set ending psw
                                                   290 ***********************************************************************
                               000003CE  00000001  291 ENDTEST  EQU   *
000003CE  5810 8E00                      00001000  292          L     R1,FAILED         did a test fail?
000003D2  1211                                     293          LTR   R1,R1
000003D4  4780 8320                      00000520  294          BZ    EOJ                  No, exit
000003D8  47F0 8338                      00000538  295          B     FAILTEST             Yes, exit with BAD PSW
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                    297 ***********************************************************************
                                                    298 *        RPTERROR                          Report instruction test in error
                                                    299 ***********************************************************************

000003DC  50F0 8240                      00000440   301 RPTERROR ST    R15,RPTSAVE          Save return address
000003E0  5050 8244                      00000444   302          ST    R5,RPTSVR5           Save R5
                                                    303 *
000003E4  4820 5004                      00000004   304          LH    R2,TNUM              get test number and convert
000003E8  4E20 8E73                      00001073   305          CVD   R2,DECNUM
000003EC  D211 8E5D 8E47       0000105D  00001047   306          MVC   PRT3,EDIT
000003F2  DE11 8E5D 8E73       0000105D  00001073   307          ED    PRT3,DECNUM
000003F8  D202 8E18 8E6A       00001018  0000106A   308          MVC   PRTNUM(3),PRT3+13    fill in message with test #
                                                    309
000003FE  D207 8E33 5008       00001033  00000008   310          MVC   PRTNAME,OPNAME       fill in message with instruction
                                                    311 *
00000404  E320 5007 0076                 00000007   312          LB    R2,m4                get m4 and convert
0000040A  4E20 8E73                      00001073   313          CVD   R2,DECNUM
0000040E  D211 8E5D 8E47       0000105D  00001047   314          MVC   PRT3,EDIT
00000414  DE11 8E5D 8E73       0000105D  00001073   315          ED    PRT3,DECNUM
0000041A  D201 8E44 8E6B       00001044  0000106B   316          MVC   PRTM4(2),PRT3+14     fill in message with m4 field

                                                    318 *
                                                    319 *        Use Hercules Diagnose for Message to console
                                                    320 *
00000420  9002 8248                      00000448   321          STM   R0,R2,RPTDWSAV       save regs used by MSG
00000424  4100 003F                      0000003F   322          LA    R0,PRTLNG            message length
00000428  4110 8E08                      00001008   323          LA    R1,PRTLINE           messagfe address
0000042C  4520 8258                      00000458   324          BAL   R2,MSG               call Hercules console MSG display
00000430  9802 8248                      00000448   325          LM    R0,R2,RPTDWSAV       restore regs


00000434  5850 8244                      00000444   327          L     R5,RPTSVR5         Restore R5
00000438  58F0 8240                      00000440   328          L     R15,RPTSAVE        Restore return address
0000043C  07FF                                      329          BR    R15                Return to caller

00000440  00000000                                  331 RPTSAVE  DC    F'0'               R15 save area
00000444  00000000                                  332 RPTSVR5  DC    F'0'               R5 save area

00000448  00000000 00000000                         334 RPTDWSAV DC    2D'0'              R0-R2 save area for MSG call
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   336 ***********************************************************************
                                                   337 *        Issue HERCULES MESSAGE pointed to by R1, length in R0
                                                   338 *              R2 = return address
                                                   339 ***********************************************************************

00000458  4900 8354                      00000554  341 MSG      CH    R0,=H'0'               Do we even HAVE a message?
0000045C  07D2                                     342          BNHR  R2                     No, ignore

0000045E  9002 8294                      00000494  344          STM   R0,R2,MSGSAVE          Save registers

00000462  4900 8356                      00000556  346          CH    R0,=AL2(L'MSGMSG)      Message length within limits?
00000466  47D0 826E                      0000046E  347          BNH   MSGOK                  Yes, continue
0000046A  4100 005F                      0000005F  348          LA    R0,L'MSGMSG            No, set to maximum

0000046E  1820                                     350 MSGOK    LR    R2,R0                  Copy length to work register
00000470  0620                                     351          BCTR  R2,0                   Minus-1 for execute
00000472  4420 82A0                      000004A0  352          EX    R2,MSGMVC              Copy message to O/P buffer

00000476  4120 200A                      0000000A  354          LA    R2,1+L'MSGCMD(,R2)     Calculate true command length
0000047A  4110 82A6                      000004A6  355          LA    R1,MSGCMD              Point to true command

0000047E  83120008                                 357          DC    X'83',X'12',X'0008'    Issue Hercules Diagnose X'008'
00000482  4780 828E                      0000048E  358          BZ    MSGRET                 Return if successful
                                                   359
00000486  1222                                     360          LTR   R2,R2                  Is Diag8 Ry (R2) 0?
00000488  4780 828E                      0000048E  361          BZ    MSGRET                   an error occurred but coninue
                                                   362
0000048C  0000                                     363          DC    H'0'                   CRASH for debugging purposes

0000048E  9802 8294                      00000494  365 MSGRET   LM    R0,R2,MSGSAVE          Restore registers
00000492  07F2                                     366          BR    R2                     Return to caller




00000494  00000000 00000000                        368 MSGSAVE  DC    3F'0'                  Registers save area
000004A0  D200 82AF 1000       000004AF  00000000  369 MSGMVC   MVC   MSGMSG(0),0(R1)        Executed instruction


000004A6  D4E2C7D5 D6C8405C                        371 MSGCMD   DC    C'MSGNOH * '           *** HERCULES MESSAGE COMMAND ***
000004AF  40404040 40404040                        372 MSGMSG   DC    CL95' '                The message text to be displayed
                                                   373
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   375 ***********************************************************************
                                                   376 *        Normal completion or Abnormal termination PSWs
                                                   377 ***********************************************************************

00000510  00020001 80000000                        379 EOJPSW   DC    0D'0',X'0002000180000000',AD(0)

00000520  B2B2 8310                     00000510   381 EOJ      LPSWE EOJPSW                Normal completion

00000528  00020001 80000000                        383 FAILPSW  DC    0D'0',X'0002000180000000',AD(X'BAD')

00000538  B2B2 8328                     00000528   385 FAILTEST LPSWE FAILPSW               Abnormal termination

                                                   387 ***********************************************************************
                                                   388 *        Working Storage
                                                   389 ***********************************************************************

0000053C  00000000                                 391 CTLR0    DS    F                 CR0
00000540  00000000                                 392          DS    F

00000544                                           394          LTORG ,                 Literals pool
00000544  00000040                                 395                =F'64'
00000548  00000002                                 396                =F'2'
0000054C  00003168                                 397                =A(E7TESTS)
00000550  00000001                                 398                =F'1'
00000554  0000                                     399                =H'0'
00000556  005F                                     400                =AL2(L'MSGMSG)
                                                   401
                                                   402 *        some constants
                                                   403
                              00000400  00000001   404 K        EQU   1024              One KB
                              00001000  00000001   405 PAGE     EQU   (4*K)             Size of one page
                              00010000  00000001   406 K64      EQU   (64*K)            64 KB
                              00100000  00000001   407 MB       EQU   (K*K)              1 MB
                                                   408
                              AABBCCDD  00000001   409 REG2PATT EQU   X'AABBCCDD'     Polluted Register pattern
                              000000DD  00000001   410 REG2LOW  EQU         X'DD'     (last byte above)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   412 *=====================================================================
                                                   413 *
                                                   414 *  NOTE: start data on an address that is easy to display
                                                   415 *        within Hercules
                                                   416 *
                                                   417 *=====================================================================
                                                   418
00000558                     00000558  00001000   419          ORG   ZVE7TST+X'1000'
00001000  00000000                                420 FAILED   DC    F'0'                     some test failed?
00001004  00000000                                421 TESTING  DC    F'0'                     current test number

                                                   423 *
                                                   424 *        failed message and associated editting
                                                   425 *
00001008  40404040 40404040                       426 PRTLINE  DC    C'         Test # '
00001018  A7A7A7                                  427 PRTNUM   DC    C'xxx'
0000101B  40868189 93858440                       428          DC    c' failed for instruction '
00001033  A7A7A7A7 A7A7A7A7                       429 PRTNAME  DC    CL8'xxxxxxxx'
0000103B  40A689A3 884094F4                       430          DC    C' with m4='
00001044  A7A7                                    431 PRTm4    DC    C'xx'
00001046  4B                                      432          DC    C'.'
                             0000003F  00000001   433 PRTLNG   EQU   *-PRTLINE
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   435 ************************************************************************
                                                   436 *        TEST failed : message working storge
                                                   437 ************************************************************************
00001047  40212020 20202020                        438 EDIT     DC    XL18'402120202020202020202020202020202020'
                                                   439
00001059  7E7E7E6E                                 440          DC    C'===>'
0000105D  40404040 40404040                        441 PRT3     DC    CL18' '
0000106F  4C7E7E7E                                 442          DC    C'<==='
00001073  00000000 00000000                        443 DECNUM   DS    CL16

                                                   445 ************************************************************************
                                                   446 *        Vector instruction results, pollution and input
                                                   447 ************************************************************************
00001084                                           448          DS    0F
00001084  00000000 00000000                        449          DS    XL16                                        gap
00001094  FFFFFFFF FFFFFFFF                        450 V1FUDGE  DC    XL16'FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF'    V1 FUDGE
000010A4  00000000 00000000                        451          DS    XL16
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   453 ***********************************************************************
                                                   454 *        E7TEST DSECT
                                                   455 ***********************************************************************

                                                   457 E7TEST   DSECT ,
00000000  00000000                                 458 TSUB     DC    A(0)           pointer  to test
00000004  0000                                     459 TNUM     DC    H'00'          Test Number
00000006  00                                       460          DC    X'00'
00000007  00                                       461 M4       DC    HL1'00'        m4 used
                                                   462
00000008  40404040 40404040                        463 OPNAME   DC    CL8' '         E6 name
00000010  00000000                                 464 V2ADDR   DC    A(0)           address of v2 source
00000014  00000000                                 465 V3ADDR   DC    A(0)           address of v3 source
00000018  00000000                                 466 RELEN    DC    A(0)           RESULT LENGTH
0000001C  00000000                                 467 READDR   DC    A(0)           result (expected) address
00000020  00000000 00000000                        468          DS    FD                gap
00000028  00000000 00000000                        469 V1OUTPUT DS    XL16           V1 Output
00000038  00000000 00000000                        470          DS    FD                gap
                                                   471
                                                   472 *        test routine will be here (from VRR-c macro)
                                                   473 *
                                                   474 *        followed by
                                                   475 *              EXPECTED RESULT

                                 00000000  00003253   477 ZVE7TST  CSECT ,
000010B4                                           478          DS    0F

                                                   480 ***********************************************************************
                                                   481 *     Macros to help build test tables
                                                   482 ***********************************************************************

                                                   484 *
                                                   485 * macro to generate individual test
                                                   486 *
                                                   487          MACRO
                                                   488          VRR_C &INST,&M4
                                                   489 .*                                &INST   - VRR-c instruction under test
                                                   490 .*                                &m4     - m3 field
                                                   491
                                                   492          GBLA  &TNUM
                                                   493 &TNUM    SETA  &TNUM+1
                                                   494
                                                   495          DS    0FD
                                                   496          USING *,R5              base for test data and test routine
                                                   497
                                                   498 T&TNUM   DC    A(X&TNUM)         address of test routine
                                                   499          DC    H'&TNUM'          test number
                                                   500          DC    X'00'
                                                   501          DC    HL1'&M4'          m4
                                                   502          DC    CL8'&INST'        instruction name
                                                   503          DC    A(RE&TNUM+16)     address of v2 source
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   504          DC    A(RE&TNUM+32)     address of v3 source
                                                   505          DC    A(16)             result length
                                                   506 REA&TNUM DC    A(RE&TNUM)        result address
                                                   507          DS    FD                gap
                                                   508 V1O&TNUM DS    XL16              V1 output
                                                   509          DS    FD                gap
                                                   510 .*
                                                   511 *
                                                   512 X&TNUM   DS    0F
                                                   513          LGF   R1,V2ADDR         load v2 source
                                                   514          VL    v22,0(R1)         use v22 to test decoder
                                                   515
                                                   516          LGF   R1,V3ADDR         load v3 source
                                                   517          VL    v23,0(R1)         use v23 to test decoder
                                                   518
                                                   519          &INST V22,V22,V23,&M4    test instruction (dest is a source)
                                                   520          VST   V22,V1O&TNUM       save v1 output
                                                   521
                                                   522          BR    R11               return
                                                   523
                                                   524 RE&TNUM  DC    0F                xl16 expected result
                                                   525
                                                   526          DROP  R5
                                                   527          MEND


                                                   529 *
                                                   530 * macro to generate table of pointers to individual tests
                                                   531 *
                                                   532          MACRO
                                                   533          PTTABLE
                                                   534          GBLA  &TNUM
                                                   535          LCLA  &CUR
                                                   536 &CUR     SETA  1
                                                   537 .*
                                                   538 TTABLE   DS    0F
                                                   539 .LOOP    ANOP
                                                   540 .*
                                                   541          DC    A(T&CUR)          TEST &CUR
                                                   542 .*
                                                   543 &CUR     SETA  &CUR+1
                                                   544          AIF   (&CUR LE &TNUM).LOOP
                                                   545 *
                                                   546          DC    A(0)              END OF TABLE
                                                   547          DC    A(0)
                                                   548 .*
                                                   549          MEND
                                                   550
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   552 ***********************************************************************
                                                   553 *        E6 VRR-c tests
                                                   554 ***********************************************************************
                                                   555          PRINT DATA
                                                   556
                                                   557 *        E7F0 VAVGL  - VECTOR AVERAGE LOGICAL
                                                   558 *        E7F2 VAVG   - VECTOR AVERAGE
                                                   559 *        E7FC VMNL   - VECTOR MINIMUM LOGICAL
                                                   560 *        E7FD VMXL   - VECTOR MAXIMUM LOGICAL
                                                   561 *        E7FE VMN    - VECTOR MINIMUM
                                                   562 *        E7FF VMX    - VECTOR MAXIMUM
                                                   563
                                                   564 *        VRR-c instruction, m4
                                                   565 *              followed by
                                                   566 *                 16 byte expected result (V1)
                                                   567 *                 16 byte V2 source
                                                   568 *                 16 byte V3 source
                                                   569 *---------------------------------------------------------------------
                                                   570 *  VMX    - VECTOR MAXIMUM
                                                   571 *---------------------------------------------------------------------
                                                   572 * Byte
                                                   573          VRR_C VMX,0
000010B8                                           574+         DS    0FD
000010B8                     000010B8              575+         USING *,R5              base for test data and test routine
000010B8  000010F8                                 576+T1       DC    A(X1)             address of test routine
000010BC  0001                                     577+         DC    H'1'              test number
000010BE  00                                       578+         DC    X'00'
000010BF  00                                       579+         DC    HL1'0'            m4
000010C0  E5D4E740 40404040                        580+         DC    CL8'VMX'          instruction name
000010C8  00001130                                 581+         DC    A(RE1+16)         address of v2 source
000010CC  00001140                                 582+         DC    A(RE1+32)         address of v3 source
000010D0  00000010                                 583+         DC    A(16)             result length
000010D4  00001120                                 584+REA1     DC    A(RE1)            result address
000010D8  00000000 00000000                        585+         DS    FD                gap
000010E0  00000000 00000000                        586+V1O1     DS    XL16              V1 output
000010E8  00000000 00000000
000010F0  00000000 00000000                        587+         DS    FD                gap
                                                   588+*
000010F8                                           589+X1       DS    0F
000010F8  E310 5010 0014            00000010       590+         LGF   R1,V2ADDR         load v2 source
000010FE  E761 0000 0806            00000000       591+         VL    v22,0(R1)         use v22 to test decoder
00001104  E310 5014 0014            00000014       592+         LGF   R1,V3ADDR         load v3 source
0000110A  E771 0000 0806            00000000       593+         VL    v23,0(R1)         use v23 to test decoder
00001110  E766 7000 0EFF                           594+         VMX   V22,V22,V23,0      test instruction (dest is a source)
00001116  E760 5028 080E            000010E0       595+         VST   V22,V1O1           save v1 output
0000111C  07FB                                     596+         BR    R11               return
00001120                                           597+RE1      DC    0F                xl16 expected result
00001120                                           598+         DROP  R5
00001120  02030405 09010181                        599          DC    XL16'020304050901018107OFFFFE00000020'   expected result
00001128  070FFFFE 00000020
00001130  01020304 09800181                        600          DC    XL16'01020304098001810 70FFFFD0000001F'   v2
00001138  070FFFFD 0000001F
00001140  02030405 0001FF80                        601          DC    XL16'020304050001FF80010AFEFE00000020'   v3
00001148  010AFEFE 00000020
                                                   602
                                                   603 * Halfword
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   604          VRR_C VMX,1
00001150                                           605+         DS    0FD
00001150                     00001150              606+         USING *,R5              base for test data and test routine
00001150  00001190                                 607+T2       DC    A(X2)             address of test routine
00001154  0002                                     608+         DC    H'2'              test number
00001156  00                                       609+         DC    X'00'
00001157  01                                       610+         DC    HL1'1'            m4
00001158  E5D4E740 40404040                        611+         DC    CL8'VMX'          instruction name
00001160  000011C8                                 612+         DC    A(RE2+16)         address of v2 source
00001164  000011D8                                 613+         DC    A(RE2+32)         address of v3 source
00001168  00000010                                 614+         DC    A(16)             result length
0000116C  000011B8                                 615+REA2     DC    A(RE2)            result address
00001170  00000000 00000000                        616+         DS    FD                gap
00001178  00000000 00000000                        617+V1O2     DS    XL16              V1 output
00001180  00000000 00000000
00001188  00000000 00000000                        618+         DS    FD                gap
                                                   619+*
00001190                                           620+X2       DS    0F
00001190  E310 5010 0014              00000010     621+         LGF   R1,V2ADDR         load v2 source
00001196  E761 0000 0806              00000000     622+         VL    v22,0(R1)         use v22 to test decoder
0000119C  E310 5014 0014              00000014     623+         LGF   R1,V3ADDR         load v3 source
000011A2  E771 0000 0806              00000000     624+         VL    v23,0(R1)         use v23 to test decoder
000011A8  E766 7000 1EFF                           625+         VMX   V22,V22,V23,1      test instruction (dest is a source)
000011AE  E760 5028 080E              00001178     626+         VST   V22,V1O2           save v1 output
000011B4  07FB                                     627+         BR    R11               return
000011B8                                           628+RE2      DC    0F                xl16 expected result
000011B8                                           629+         DROP  R5
000011B8  00020001 FFFE0001                        630          DC    XL16'00020001FFFE00017FFF800112340020'   expected result
000011C0  7FFF8001 12340020
000011C8  0001FFFF FFFD8000                        631          DC    XL16'0001FFFFFFFD80007FFF80000123001F'   v2
000011D0  7FFF8000 0123001F
000011D8  00020001 FFFE0001                        632          DC    XL16'00020001FFFE000100AA800112340020'   v3
000011E0  00AA8001 12340020
                                                   633
                                                   634 * Word
                                                   635          VRR_C VMX,2
000011E8                                           636+         DS    0FD
000011E8                     000011E8              637+         USING *,R5              base for test data and test routine
000011E8  00001228                                 638+T3       DC    A(X3)             address of test routine
000011EC  0003                                     639+         DC    H'3'              test number
000011EE  00                                       640+         DC    X'00'
000011EF  02                                       641+         DC    HL1'2'            m4
000011F0  E5D4E740 40404040                        642+         DC    CL8'VMX'          instruction name
000011F8  00001260                                 643+         DC    A(RE3+16)         address of v2 source
000011FC  00001270                                 644+         DC    A(RE3+32)         address of v3 source
00001200  00000010                                 645+         DC    A(16)             result length
00001204  00001250                                 646+REA3     DC    A(RE3)            result address
00001208  00000000 00000000                        647+         DS    FD                gap
00001210  00000000 00000000                        648+V1O3     DS    XL16              V1 output
00001218  00000000 00000000
00001220  00000000 00000000                        649+         DS    FD                gap
                                                   650+*
00001228                                           651+X3       DS    0F
00001228  E310 5010 0014              00000010     652+         LGF   R1,V2ADDR         load v2 source
0000122E  E761 0000 0806              00000000     653+         VL    v22,0(R1)         use v22 to test decoder
00001234  E310 5014 0014              00000014     654+         LGF   R1,V3ADDR         load v3 source
```

```
  LOC       OBJECT CODE      ADDR1     ADDR2    STMT

0000123A  E771 0000 0806               00000000   655+          VL    v23,0(R1)         use v23 to test decoder
00001240  E766 7000 2EFF                          656+          VMX   V22,V22,V23,2      test instruction (dest is a source)
00001246  E760 9010 080E               00001210   657+          VST   V22,V1O3           save v1 output
0000124C  07FB                                    658+          BR    R11               return
00001250                                          659+RE3       DC    0F                xl16 expected result
00001250                                          660+          DROP  R5
00001250  FFFFFFFF 7FFFFFFF                       661           DC    XL16'FFFFFFFF7FFFFFFF1234567800000020'   expected result
00001258  12345678 00000020
00001260  FFFFFFFF 7FFFFFFF                       662           DC    XL16'FFFFFFFF7FFFFFFF012345670000001F'   v2
00001268  01234567 0000001F
00001270  FFFFFFFE 0000000A                       663           DC    XL16'FFFFFFFE0000000A1234567800000020'   v3
00001278  12345678 00000020
                                                  664
                                                  665 * Doubleword
                                                  666           VRR_C VMX,3
00001280                                          667+          DS    0FD
00001280                     00001280             668+          USING *,R5              base for test data and test routine
00001280  000012C0                                669+T4        DC    A(X4)             address of test routine
00001284  0004                                    670+          DC    H'4'              test number
00001286  00                                      671+          DC    X'00'
00001287  03                                      672+          DC    HL1'3'            m4
00001288  E5D4E740 40404040                       673+          DC    CL8'VMX'          instruction name
00001290  000012F8                                674+          DC    A(RE4+16)         address of v2 source
00001294  00001308                                675+          DC    A(RE4+32)         address of v3 source
00001298  00000010                                676+          DC    A(16)             result length
0000129C  000012E8                                677+REA4      DC    A(RE4)            result address
000012A0  00000000 00000000                       678+          DS    FD                gap
000012A8  00000000 00000000                       679+V1O4      DS    XL16              V1 output
000012B0  00000000 00000000
000012B8  00000000 00000000                       680+          DS    FD                gap
                                                  681+*
000012C0                                          682+X4        DS    0F
000012C0  E310 5010 0014               00000010   683+          LGF   R1,V2ADDR         load v2 source
000012C6  E761 0000 0806               00000000   684+          VL    v22,0(R1)         use v22 to test decoder
000012CC  E310 5014 0014               00000014   685+          LGF   R1,V3ADDR         load v3 source
000012D2  E771 0000 0806               00000000   686+          VL    v23,0(R1)         use v23 to test decoder
000012D8  E766 7000 3EFF                          687+          VMX   V22,V22,V23,3      test instruction (dest is a source)
000012DE  E760 5028 080E               000012A8   688+          VST   V22,V1O4           save v1 output
000012E4  07FB                                    689+          BR    R11               return
000012E8                                          690+RE4       DC    0F                xl16 expected result
000012E8                                          691+          DROP  R5
000012E8  FFFFFFFF FFFFFFFF                       692           DC    XL16'FFFFFFFFFFFFFFFF0000000000000020'   expected result
000012F0  00000000 00000020
000012F8  FFFFFFFF FFFFFFFF                       693           DC    XL16'FFFFFFFFFFFFFFFF000000000000001F'   v2
00001300  00000000 0000001F
00001308  FFFFFFFF FFFFFFFD                       694           DC    XL16'FFFFFFFFFFFFFFFD0000000000000020'   v3
00001310  00000000 00000020
                                                  695
                                                  696 * quadword
                                                  697           VRR_C VMX,4
00001318                                          698+          DS    0FD
00001318                     00001318             699+          USING *,R5              base for test data and test routine
00001318  00001358                                700+T5        DC    A(X5)             address of test routine
0000131C  0005                                    701+          DC    H'5'              test number
0000131E  00                                      702+          DC    X'00'
0000131F  04                                      703+          DC    HL1'4'            m4
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00001320  E5D4E740 40404040                        704+          DC    CL8'VMX'           instruction name
00001328  00001390                                 705+          DC    A(RE5+16)          address of v2 source
0000132C  000013A0                                 706+          DC    A(RE5+32)          address of v3 source
00001330  00000010                                 707+          DC    A(16)              result length
00001334  00001380                                 708+REA5      DC    A(RE5)             result address
00001338  00000000 00000000                        709+          DS    FD                 gap
00001340  00000000 00000000                        710+V1O5      DS    XL16               V1 output
00001348  00000000 00000000
00001350  00000000 00000000                        711+          DS    FD                 gap
                                                   712+*
00001358                                           713+X5        DS    0F
00001358  E310 5010 0014              00000010     714+          LGF   R1,V2ADDR          load v2 source
0000135E  E761 0000 0806              00000000     715+          VL    v22,0(R1)          use v22 to test decoder
00001364  E310 5014 0014              00000014     716+          LGF   R1,V3ADDR          load v3 source
0000136A  E771 0000 0806              00000000     717+          VL    v23,0(R1)          use v23 to test decoder
00001370  E766 7000 4EFF                           718+          VMX   V22,V22,V23,4       test instruction (dest is a source)
00001376  E760 5028 080E              00001340     719+          VST   V22,V1O5            save v1 output
0000137C  07FB                                     720+          BR    R11                return
00001380                                           721+RE5       DC    0F                 xl16 expected result
00001380                                           722+          DROP  R5
00001380  FFFFFFFF FFFFFFFF                        723           DC    XL16'FFFFFFFFFFFFFFFF000000000000001F'   expected result
00001388  00000000 0000001F
00001390  FFFFFFFF FFFFFFFF                        724           DC    XL16'FFFFFFFFFFFFFFFF000000000000001F'   v2
00001398  00000000 0000001F
000013A0  FFFFFFFF FFFFFFFD                        725           DC    XL16'FFFFFFFFFFFFFFFD0000000000000020'   v3
000013A8  00000000 00000020
                                                   726
                                                   727 * quadword
                                                   728           VRR_C VMX,4
000013B0                                           729+          DS    0FD
000013B0                     000013B0              730+          USING *,R5               base for test data and test routine
000013B0  000013F0                                 731+T6        DC    A(X6)              address of test routine
000013B4  0006                                     732+          DC    H'6'               test number
000013B6  00                                       733+          DC    X'00'
000013B7  04                                       734+          DC    HL1'4'             m4
000013B8  E5D4E740 40404040                        735+          DC    CL8'VMX'           instruction name
000013C0  00001428                                 736+          DC    A(RE6+16)          address of v2 source
000013C4  00001438                                 737+          DC    A(RE6+32)          address of v3 source
000013C8  00000010                                 738+          DC    A(16)              result length
000013CC  00001418                                 739+REA6      DC    A(RE6)             result address
000013D0  00000000 00000000                        740+          DS    FD                 gap
000013D8  00000000 00000000                        741+V1O6      DS    XL16               V1 output
000013E0  00000000 00000000
000013E8  00000000 00000000                        742+          DS    FD                 gap
                                                   743+*
000013F0                                           744+X6        DS    0F
000013F0  E310 5010 0014              00000010     745+          LGF   R1,V2ADDR          load v2 source
000013F6  E761 0000 0806              00000000     746+          VL    v22,0(R1)          use v22 to test decoder
000013FC  E310 5014 0014              00000014     747+          LGF   R1,V3ADDR          load v3 source
00001402  E771 0000 0806              00000000     748+          VL    v23,0(R1)          use v23 to test decoder
00001408  E766 7000 4EFF                           749+          VMX   V22,V22,V23,4       test instruction (dest is a source)
0000140E  E760 5028 080E              000013D8     750+          VST   V22,V1O6            save v1 output
00001414  07FB                                     751+          BR    R11                return
00001418                                           752+RE6       DC    0F                 xl16 expected result
00001418                                           753+          DROP  R5
00001418  0001FFFF FFFD8000                        754           DC    XL16'0001FFFFFFFD80007FFF80000123001F'  expected result
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00001420  7FFF8000 0123001F
00001428  0001FFFF FFFD8000                        755          DC    XL16'0001FFFFFFFD80007FFF80000123001F'   v2
00001430  7FFF8000 0123001F
00001438  FFFFFFFF FFFFFFFD                        756          DC    XL16'FFFFFFFFFFFFFFFD0000000000000020'   v3
00001440  00000000 00000020
                                                   757
                                                   758 * quadword
                                                   759          VRR_C VMX,4
00001448                                           760+         DS    0FD
00001448                      00001448             761+         USING *,R5              base for test data and test routine
00001448  00001488                                 762+T7       DC    A(X7)             address of test routine
0000144C  0007                                     763+         DC    H'7'              test number
0000144E  00                                       764+         DC    X'00'
0000144F  04                                       765+         DC    HL1'4'            m4
00001450  E5D4E740 40404040                        766+         DC    CL8'VMX'          instruction name
00001458  000014C0                                 767+         DC    A(RE7+16)         address of v2 source
0000145C  000014D0                                 768+         DC    A(RE7+32)         address of v3 source
00001460  00000010                                 769+         DC    A(16)             result length
00001464  000014B0                                 770+REA7     DC    A(RE7)            result address
00001468  00000000 00000000                        771+         DS    FD                gap
00001470  00000000 00000000                        772+V1O7     DS    XL16              V1 output
00001478  00000000 00000000
00001480  00000000 00000000                        773+         DS    FD                gap
                                                   774+*
00001488                                           775+X7       DS    0F
00001488  E310 5010 0014               00000010    776+         LGF   R1,V2ADDR         load v2 source
0000148E  E761 0000 0806               00000000    777+         VL    v22,0(R1)         use v22 to test decoder
00001494  E310 5014 0014               00000014    778+         LGF   R1,V3ADDR         load v3 source
0000149A  E771 0000 0806               00000000    779+         VL    v23,0(R1)         use v23 to test decoder
000014A0  E766 7000 4EFF                           780+         VMX   V22,V22,V23,4      test instruction (dest is a source)
000014A6  E760 5028 080E               00001470    781+         VST   V22,V1O7           save v1 output
000014AC  07FB                                     782+         BR    R11               return
000014B0                                           783+RE7      DC    0F                xl16 expected result
000014B0                                           784+         DROP  R5
000014B0  00020001 FFFE0001                        785          DC    XL16'00020001FFFE000100AA800112340020'   expected result
000014B8  00AA8001 12340020
000014C0  0001FFFF FFFD8000                        786          DC    XL16'0001FFFFFFFD80007FFF80000123001F'   v2
000014C8  7FFF8000 0123001F
000014D0  00020001 FFFE0001                        787          DC    XL16'00020001FFFE000100AA800112340020'   v3
000014D8  00AA8001 12340020
                                                   788
                                                   789 *--------------------------------------------------------------------
                                                   790 * VMXL   - VECTOR MAXIMUM LOGICAL
                                                   791 *--------------------------------------------------------------------
                                                   792 * Byte
                                                   793          VRR_C VMXL,0
000014E0                                           794+         DS    0FD
000014E0                      000014E0             795+         USING *,R5              base for test data and test routine
000014E0  00001520                                 796+T8       DC    A(X8)             address of test routine
000014E4  0008                                     797+         DC    H'8'              test number
000014E6  00                                       798+         DC    X'00'
000014E7  00                                       799+         DC    HL1'0'            m4
000014E8  E5D4E7D3 40404040                        800+         DC    CL8'VMXL'         instruction name
000014F0  00001558                                 801+         DC    A(RE8+16)         address of v2 source
000014F4  00001568                                 802+         DC    A(RE8+32)         address of v3 source
000014F8  00000010                                 803+         DC    A(16)             result length
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

000014FC  00001548                                 804+REA8      DC    A(RE8)              result address
00001500  00000000 00000000                        805+          DS    FD                  gap
00001508  00000000 00000000                        806+V108      DS    XL16                V1 output
00001510  00000000 00000000
00001518  00000000 00000000                        807+          DS    FD                  gap
                                                   808+*
00001520                                           809+X8        DS    0F
00001520  E310 5010 0014              00000010     810+          LGF   R1,V2ADDR           load v2 source
00001526  E761 0000 0806              00000000     811+          VL    v22,0(R1)           use v22 to test decoder
0000152C  E310 5014 0014              00000014     812+          LGF   R1,V3ADDR           load v3 source
00001532  E771 0000 0806              00000000     813+          VL    v23,0(R1)           use v23 to test decoder
00001538  E766 7000 0EFD                           814+          VMXL  V22,V22,V23,0        test instruction (dest is a source)
0000153E  E760 5028 080E              00001508     815+          VST   V22,V108             save v1 output
00001544  07FB                                     816+          BR    R11                 return
00001548                                           817+RE8       DC    0F                  xl16 expected result
00001548                                           818+          DROP  R5
00001548  02030405 0980FF81                        819           DC    XL16'020304050980FF81070FFFFE00000020'   expected result
00001550  070FFFFE 00000020
00001558  01020304 09800181                        820           DC    XL16'0102030409800181070FFFFD0000001F'   v2
00001560  070FFFFD 0000001F
00001568  02030405 0001FF80                        821           DC    XL16'020304050001FF80010AFEFE00000020'   v3
00001570  010AFEFE 00000020
                                                   822
                                                   823 * Halfword
                                                   824           VRR_C VMXL,1
00001578                                           825+          DS    0FD
00001578                        00001578           826+          USING *,R5                base for test data and test routine
00001578  000015B8                                 827+T9        DC    A(X9)               address of test routine
0000157C  0009                                     828+          DC    H'9'                test number
0000157E  00                                       829+          DC    X'00'
0000157F  01                                       830+          DC    HL1'1'              m4
00001580  E5D4E7D3 40404040                        831+          DC    CL8'VMXL'           instruction name
00001588  000015F0                                 832+          DC    A(RE9+16)           address of v2 source
0000158C  00001600                                 833+          DC    A(RE9+32)           address of v3 source
00001590  00000010                                 834+          DC    A(16)               result length
00001594  000015E0                                 835+REA9      DC    A(RE9)              result address
00001598  00000000 00000000                        836+          DS    FD                  gap
000015A0  00000000 00000000                        837+V109      DS    XL16                V1 output
000015A8  00000000 00000000
000015B0  00000000 00000000                        838+          DS    FD                  gap
                                                   839+*
000015B8                                           840+X9        DS    0F
000015B8  E310 5010 0014              00000010     841+          LGF   R1,V2ADDR           load v2 source
000015BE  E761 0000 0806              00000000     842+          VL    v22,0(R1)           use v22 to test decoder
000015C4  E310 5014 0014              00000014     843+          LGF   R1,V3ADDR           load v3 source
000015CA  E771 0000 0806              00000000     844+          VL    v23,0(R1)           use v23 to test decoder
000015D0  E766 7000 1EFD                           845+          VMXL  V22,V22,V23,1        test instruction (dest is a source)
000015D6  E760 5028 080E              000015A0     846+          VST   V22,V109             save v1 output
000015DC  07FB                                     847+          BR    R11                 return
000015E0                                           848+RE9       DC    0F                  xl16 expected result
000015E0                                           849+          DROP  R5
000015E0  0002FFFF FFFE8000                        850           DC    XL16'0002FFFFFFFE80007FFF800112340020'   expected result
000015E8  7FFF8001 12340020
000015F0  0001FFFF FFFD8000                        851           DC    XL16'0001FFFFFFFD80007FFF80000123001F'   v2
000015F8  7FFF8000 0123001F
00001600  00020001 FFFE0001                        852           DC    XL16'00020001FFFE000100AA800112340020'   v3
```

```
  LOC       OBJECT CODE       ADDR1     ADDR2    STMT

00001608  00AA8001 12340020
                                                  853
                                                  854 * Word
                                                  855          VRR_C VMXL,2
00001610                                          856+         DS    0FD
00001610                      00001610            857+         USING *,R5              base for test data and test routine
00001610  00001650                                858+T10      DC    A(X10)            address of test routine
00001614  000A                                    859+         DC    H'10'             test number
00001616  00                                      860+         DC    X'00'
00001617  02                                      861+         DC    HL1'2'            m4
00001618  E5D4E7D3 40404040                       862+         DC    CL8'VMXL'         instruction name
00001620  00001688                                863+         DC    A(RE10+16)        address of v2 source
00001624  00001698                                864+         DC    A(RE10+32)        address of v3 source
00001628  00000010                                865+         DC    A(16)             result length
0000162C  00001678                                866+REA10    DC    A(RE10)           result address
00001630  00000000 00000000                       867+         DS    FD                gap
00001638  00000000 00000000                       868+V1O10    DS    XL16              V1 output
00001640  00000000 00000000
00001648  00000000 00000000                       869+         DS    FD                gap
                                                  870+*
00001650                                          871+X10      DS    0F
00001650  E310 5010 0014             00000010     872+         LGF   R1,V2ADDR         load v2 source
00001656  E761 0000 0806             00000000     873+         VL    v22,0(R1)         use v22 to test decoder
0000165C  E310 5014 0014             00000014     874+         LGF   R1,V3ADDR         load v3 source
00001662  E771 0000 0806             00000000     875+         VL    v23,0(R1)         use v23 to test decoder
00001668  E766 7000 2EFD                          876+         VMXL  V22,V22,V23,2      test instruction (dest is a source)
0000166E  E760 5028 080E             00001638     877+         VST   V22,V1O10          save v1 output
00001674  07FB                                    878+         BR    R11               return
00001678                                          879+RE10     DC    0F                xl16 expected result
00001678                                          880+         DROP  R5
00001678  FFFFFFFF 7FFFFFFF                       881          DC    XL16'FFFFFFFF7FFFFFFF1234567800000020'   expected result
00001680  12345678 00000020
00001688  FFFFFFFF 7FFFFFFF                       882          DC    XL16'FFFFFFFF7FFFFFFF012345670000001F'   v2
00001690  01234567 0000001F
00001698  FFFFFFFE 0000000A                       883          DC    XL16'FFFFFFFE0000000A1234567800000020'   v3
000016A0  12345678 00000020
                                                  884
                                                  885 * Doubleword
                                                  886          VRR_C VMXL,3
000016A8                                          887+         DS    0FD
000016A8                      000016A8            888+         USING *,R5              base for test data and test routine
000016A8  000016E8                                889+T11      DC    A(X11)            address of test routine
000016AC  000B                                    890+         DC    H'11'             test number
000016AE  00                                      891+         DC    X'00'
000016AF  03                                      892+         DC    HL1'3'            m4
000016B0  E5D4E7D3 40404040                       893+         DC    CL8'VMXL'         instruction name
000016B8  00001720                                894+         DC    A(RE11+16)        address of v2 source
000016BC  00001730                                895+         DC    A(RE11+32)        address of v3 source
000016C0  00000010                                896+         DC    A(16)             result length
000016C4  00001710                                897+REA11    DC    A(RE11)           result address
000016C8  00000000 00000000                       898+         DS    FD                gap
000016D0  00000000 00000000                       899+V1O11    DS    XL16              V1 output
000016D8  00000000 00000000
000016E0  00000000 00000000                       900+         DS    FD                gap
                                                  901+*
000016E8                                          902+X11      DS    0F
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

000016E8  E310 5010 0014                00000010   903+          LGF   R1,V2ADDR           load v2 source
000016EE  E761 0000 0806                00000000   904+          VL    v22,0(R1)           use v22 to test decoder
000016F4  E310 5014 0014                00000014   905+          LGF   R1,V3ADDR           load v3 source
000016FA  E771 0000 0806                00000000   906+          VL    v23,0(R1)           use v23 to test decoder
00001700  E766 7000 3EFD                           907+          VMXL  V22,V22,V23,3        test instruction (dest is a source)
00001706  E760 5028 080E                000016D0   908+          VST   V22,V1O11            save v1 output
0000170C  07FB                                     909+          BR    R11                 return
00001710                                           910+RE11      DC    0F                  xl16 expected result
00001710                                           911+          DROP  R5
00001710  FFFFFFFF FFFFFFFF                        912           DC    XL16'FFFFFFFFFFFFFFFF0000000000000020'   expected result
00001718  00000000 00000020
00001720  FFFFFFFF FFFFFFFF                        913           DC    XL16'FFFFFFFFFFFFFFFF000000000000001F'   v2
00001728  00000000 0000001F
00001730  FFFFFFFF FFFFFFFD                        914           DC    XL16'FFFFFFFFFFFFFFFD0000000000000020'   v3
00001738  00000000 00000020
                                                   915
                                                   916 * quadword
                                                   917           VRR_C VMXL,4
00001740                                           918+          DS    0FD
00001740                      00001740             919+          USING *,R5                base for test data and test routine
00001740  00001780                                 920+T12       DC    A(X12)              address of test routine
00001744  000C                                     921+          DC    H'12'               test number
00001746  00                                       922+          DC    X'00'
00001747  04                                       923+          DC    HL1'4'              m4
00001748  E5D4E7D3 40404040                        924+          DC    CL8'VMXL'           instruction name
00001750  000017B8                                 925+          DC    A(RE12+16)          address of v2 source
00001754  000017C8                                 926+          DC    A(RE12+32)          address of v3 source
00001758  00000010                                 927+          DC    A(16)               result length
0000175C  000017A8                                 928+REA12     DC    A(RE12)             result address
00001760  00000000 00000000                        929+          DS    FD                  gap
00001768  00000000 00000000                        930+V1O12     DS    XL16                V1 output
00001770  00000000 00000000
00001778  00000000 00000000                        931+          DS    FD                  gap
                                                   932+*
00001780                                           933+X12       DS    0F
00001780  E310 5010 0014                00000010   934+          LGF   R1,V2ADDR           load v2 source
00001786  E761 0000 0806                00000000   935+          VL    v22,0(R1)           use v22 to test decoder
0000178C  E310 5014 0014                00000014   936+          LGF   R1,V3ADDR           load v3 source
00001792  E771 0000 0806                00000000   937+          VL    v23,0(R1)           use v23 to test decoder
00001798  E766 7000 4EFD                           938+          VMXL  V22,V22,V23,4        test instruction (dest is a source)
0000179E  E760 5028 080E                00001768   939+          VST   V22,V1O12            save v1 output
000017A4  07FB                                     940+          BR    R11                 return
000017A8                                           941+RE12      DC    0F                  xl16 expected result
000017A8                                           942+          DROP  R5
000017A8  FFFFFFFF FFFFFFFF                        943           DC    XL16'FFFFFFFFFFFFFFFF000000000000001F'   expected result
000017B0  00000000 0000001F
000017B8  FFFFFFFF FFFFFFFF                        944           DC    XL16'FFFFFFFFFFFFFFFF000000000000001F'   v2
000017C0  00000000 0000001F
000017C8  FFFFFFFF FFFFFFFD                        945           DC    XL16'FFFFFFFFFFFFFFFD0000000000000020'   v3
000017D0  00000000 00000020
                                                   946
                                                   947 * quadword
                                                   948           VRR_C VMXL,4
000017D8                                           949+          DS    0FD
000017D8                      000017D8             950+          USING *,R5                base for test data and test routine
000017D8  00001818                                 951+T13       DC    A(X13)              address of test routine
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

000017DC  000D                                     952+         DC    H'13'              test number
000017DE  00                                       953+         DC    X'00'
000017DF  04                                       954+         DC    HL1'4'             m4
000017E0  E5D4E7D3 40404040                        955+         DC    CL8'VMXL'          instruction name
000017E8  00001850                                 956+         DC    A(RE13+16)         address of v2 source
000017EC  00001860                                 957+         DC    A(RE13+32)         address of v3 source
000017F0  00000010                                 958+         DC    A(16)              result length
000017F4  00001840                                 959+REA13    DC    A(RE13)            result address
000017F8  00000000 00000000                        960+         DS    FD                 gap
00001800  00000000 00000000                        961+V1O13    DS    XL16               V1 output
00001808  00000000 00000000
00001810  00000000 00000000                        962+         DS    FD                 gap
                                                   963+*
00001818                                           964+X13      DS    0F
00001818  E310 5010 0014             00000010      965+         LGF   R1,V2ADDR          load v2 source
0000181E  E761 0000 0806             00000000      966+         VL    v22,0(R1)          use v22 to test decoder
00001824  E310 5014 0014             00000014      967+         LGF   R1,V3ADDR          load v3 source
0000182A  E771 0000 0806             00000000      968+         VL    v23,0(R1)          use v23 to test decoder
00001830  E766 7000 4EFD                           969+         VMXL  V22,V22,V23,4       test instruction (dest is a source)
00001836  E760 5028 080E             00001800      970+         VST   V22,V1O13           save v1 output
0000183C  07FB                                     971+         BR    R11                return
00001840                                           972+RE13     DC    0F                 xl16 expected result
00001840                                           973+         DROP  R5
00001840  FFFFFFFF FFFFFFFD                        974          DC    XL16'FFFFFFFFFFFFFFFD0000000000000020'  expected result
00001848  00000000 00000020
00001850  0001FFFF FFFD8000                        975          DC    XL16'0001FFFFFFFD80007FFF80000123001F'   v2
00001858  7FFF8000 0123001F
00001860  FFFFFFFF FFFFFFFD                        976          DC    XL16'FFFFFFFFFFFFFFFD0000000000000020'   v3
00001868  00000000 00000020
                                                   977
                                                   978 * quadword
                                                   979          VRR_C VMXL,4
00001870                                           980+         DS    0FD
00001870                           00001870        981+         USING *,R5               base for test data and test routine
00001870  000018B0                                 982+T14      DC    A(X14)             address of test routine
00001874  000E                                     983+         DC    H'14'              test number
00001876  00                                       984+         DC    X'00'
00001877  04                                       985+         DC    HL1'4'             m4
00001878  E5D4E7D3 40404040                        986+         DC    CL8'VMXL'          instruction name
00001880  000018E8                                 987+         DC    A(RE14+16)         address of v2 source
00001884  000018F8                                 988+         DC    A(RE14+32)         address of v3 source
00001888  00000010                                 989+         DC    A(16)              result length
0000188C  000018D8                                 990+REA14    DC    A(RE14)            result address
00001890  00000000 00000000                        991+         DS    FD                 gap
00001898  00000000 00000000                        992+V1O14    DS    XL16               V1 output
000018A0  00000000 00000000
000018A8  00000000 00000000                        993+         DS    FD                 gap
                                                   994+*
000018B0                                           995+X14      DS    0F
000018B0  E310 5010 0014             00000010      996+         LGF   R1,V2ADDR          load v2 source
000018B6  E761 0000 0806             00000000      997+         VL    v22,0(R1)          use v22 to test decoder
000018BC  E310 5014 0014             00000014      998+         LGF   R1,V3ADDR          load v3 source
000018C2  E771 0000 0806             00000000      999+         VL    v23,0(R1)          use v23 to test decoder
000018C8  E766 7000 4EFD                          1000+         VMXL  V22,V22,V23,4       test instruction (dest is a source)
000018CE  E760 5028 080E             00001898     1001+         VST   V22,V1O14           save v1 output
000018D4  07FB                                    1002+         BR    R11                return
```

```
  LOC       OBJECT CODE       ADDR1     ADDR2    STMT

000018D8                                         1003+RE14      DC    0F                  xl16 expected result
000018D8                                         1004+          DROP  R5
000018D8  00020001 FFFE0001                      1005           DC    XL16'00020001FFFE000100AA800112340020'   expected result
000018E0  00AA8001 12340020
000018E8  0001FFFF FFFD8000                      1006           DC    XL16'0001FFFFFFFD80007FFF80000123001F'   v2
000018F0  7FFF8000 0123001F
000018F8  00020001 FFFE0001                      1007           DC    XL16'00020001FFFE000100AA800112340020'   v3
00001900  00AA8001 12340020
                                                 1008
                                                 1009 *--------------------------------------------------------------------
                                                 1010 *  VMN    - VECTOR MINIMUM
                                                 1011 *--------------------------------------------------------------------
                                                 1012 * Byte
                                                 1013           VRR_C VMN,0
00001908                                         1014+          DS    0FD
00001908                     00001908            1015+          USING *,R5                base for test data and test routine
00001908  00001948                               1016+T15       DC    A(X15)              address of test routine
0000190C  000F                                   1017+          DC    H'15'               test number
0000190E  00                                     1018+          DC    X'00'
0000190F  00                                     1019+          DC    HL1'0'              m4
00001910  E5D4D540 40404040                      1020+          DC    CL8'VMN'            instruction name
00001918  00001980                               1021+          DC    A(RE15+16)          address of v2 source
0000191C  00001990                               1022+          DC    A(RE15+32)          address of v3 source
00001920  00000010                               1023+          DC    A(16)               result length
00001924  00001970                               1024+REA15     DC    A(RE15)             result address
00001928  00000000 00000000                      1025+          DS    FD                  gap
00001930  00000000 00000000                      1026+V1O15     DS    XL16                V1 output
00001938  00000000 00000000
00001940  00000000 00000000                      1027+          DS    FD                  gap
                                                 1028+*
00001948                                         1029+X15       DS    0F
00001948  E310 5010 0014            00000010     1030+          LGF   R1,V2ADDR           load v2 source
0000194E  E761 0000 0806            00000000     1031+          VL    v22,0(R1)           use v22 to test decoder
00001954  E310 5014 0014            00000014     1032+          LGF   R1,V3ADDR           load v3 source
0000195A  E771 0000 0806            00000000     1033+          VL    v23,0(R1)           use v23 to test decoder
00001960  E766 7000 0EFE                         1034+          VMN   V22,V22,V23,0        test instruction (dest is a source)
00001966  E760 5028 080E            00001930     1035+          VST   V22,V1O15            save v1 output
0000196C  07FB                                   1036+          BR    R11                 return
00001970                                         1037+RE15      DC    0F                  xl16 expected result
00001970                                         1038+          DROP  R5
00001970  01020304 0080FF80                      1039           DC    XL16'010203040080FF80010AFEFD0000001F'   expected result
00001978  010AFEFD 0000001F
00001980  01020304 09800181                      1040           DC    XL16'010203040980018107OFFFFD0000001F'   v2
00001988  070FFFFD 0000001F
00001990  02030405 0001FF80                      1041           DC    XL16'020304050001FF80010AFEFE00000020'   v3
00001998  010AFEFE 00000020
                                                 1042
                                                 1043 * Halfword
                                                 1044           VRR_C VMN,1
000019A0                                         1045+          DS    0FD
000019A0                     000019A0            1046+          USING *,R5                base for test data and test routine
000019A0  000019E0                               1047+T16       DC    A(X16)              address of test routine
000019A4  0010                                   1048+          DC    H'16'               test number
000019A6  00                                     1049+          DC    X'00'
000019A7  01                                     1050+          DC    HL1'1'              m4
000019A8  E5D4D540 40404040                      1051+          DC    CL8'VMN'            instruction name
```

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  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000019B0  00001A18                                 1052+         DC    A(RE16+16)         address of v2 source
000019B4  00001A28                                 1053+         DC    A(RE16+32)         address of v3 source
000019B8  00000010                                 1054+         DC    A(16)              result length
000019BC  00001A08                                 1055+REA16    DC    A(RE16)            result address
000019C0  00000000 00000000                        1056+         DS    FD                 gap
000019C8  00000000 00000000                        1057+V1O16    DS    XL16               V1 output
000019D0  00000000 00000000
000019D8  00000000 00000000                        1058+         DS    FD                 gap
                                                   1059+*
000019E0                                           1060+X16      DS    0F
000019E0  E310 5010 0014                00000010   1061+         LGF   R1,V2ADDR          load v2 source
000019E6  E761 0000 0806                00000000   1062+         VL    v22,0(R1)          use v22 to test decoder
000019EC  E310 5014 0014                00000014   1063+         LGF   R1,V3ADDR          load v3 source
000019F2  E771 0000 0806                00000000   1064+         VL    v23,0(R1)          use v23 to test decoder
000019F8  E766 7000 1EFE                           1065+         VMN   V22,V22,V23,1       test instruction (dest is a source)
000019FE  E760 5028 080E                000019C8   1066+         VST   V22,V1O16           save v1 output
00001A04  07FB                                     1067+         BR    R11                return
00001A08                                           1068+RE16     DC    0F                 xl16 expected result
00001A08                                           1069+         DROP  R5
00001A08  0001FFFF FFFD8000                        1070          DC    XL16'0001FFFFFFFD800000AA80000123001F'   expected result
00001A10  00AA8000 0123001F
00001A18  0001FFFF FFFD8000                        1071          DC    XL16'0001FFFFFFFD80007FFF80000123001F'   v2
00001A20  7FFF8000 0123001F
00001A28  00020001 FFFE0001                        1072          DC    XL16'00020001FFFE000100AA800112340020'   v3
00001A30  00AA8001 12340020
                                                   1073
                                                   1074 * Word
                                                   1075          VRR_C VMN,2
00001A38                                           1076+         DS    0FD
00001A38                     00001A38              1077+         USING *,R5               base for test data and test routine
00001A38  00001A78                                 1078+T17      DC    A(X17)             address of test routine
00001A3C  0011                                     1079+         DC    H'17'              test number
00001A3E  00                                       1080+         DC    X'00'
00001A3F  02                                       1081+         DC    HL1'2'             m4
00001A40  E5D4D540 40404040                        1082+         DC    CL8'VMN'           instruction name
00001A48  00001AB0                                 1083+         DC    A(RE17+16)         address of v2 source
00001A4C  00001AC0                                 1084+         DC    A(RE17+32)         address of v3 source
00001A50  00000010                                 1085+         DC    A(16)              result length
00001A54  00001AA0                                 1086+REA17    DC    A(RE17)            result address
00001A58  00000000 00000000                        1087+         DS    FD                 gap
00001A60  00000000 00000000                        1088+V1O17    DS    XL16               V1 output
00001A68  00000000 00000000
00001A70  00000000 00000000                        1089+         DS    FD                 gap
                                                   1090+*
00001A78                                           1091+X17      DS    0F
00001A78  E310 5010 0014                00000010   1092+         LGF   R1,V2ADDR          load v2 source
00001A7E  E761 0000 0806                00000000   1093+         VL    v22,0(R1)          use v22 to test decoder
00001A84  E310 5014 0014                00000014   1094+         LGF   R1,V3ADDR          load v3 source
00001A8A  E771 0000 0806                00000000   1095+         VL    v23,0(R1)          use v23 to test decoder
00001A90  E766 7000 2EFE                           1096+         VMN   V22,V22,V23,2       test instruction (dest is a source)
00001A96  E760 5028 080E                00001A60   1097+         VST   V22,V1O17           save v1 output
00001A9C  07FB                                     1098+         BR    R11                return
00001AA0                                           1099+RE17     DC    0F                 xl16 expected result
00001AA0                                           1100+         DROP  R5
00001AA0  FFFFFFFE 0000000A                        1101          DC    XL16'FFFFFFFE0000000A012345670000001F'   expected result
00001AA8  01234567 0000001F
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00001AB0  FFFFFFFF 7FFFFFFF                        1102           DC    XL16'FFFFFFFF7FFFFFFF012345670000001F'   v2
00001AB8  01234567 0000001F
00001AC0  FFFFFFFE 0000000A                        1103           DC    XL16'FFFFFFFE0000000A1234567800000020'   v3
00001AC8  12345678 00000020
                                                   1104
                                                   1105 * Doubleword
                                                   1106           VRR_C VMN,3
00001AD0                                           1107+          DS    0FD
00001AD0                     00001AD0              1108+          USING *,R5              base for test data and test routine
00001AD0  00001B10                                 1109+T18       DC    A(X18)            address of test routine
00001AD4  0012                                     1110+          DC    H'18'             test number
00001AD6  00                                       1111+          DC    X'00'
00001AD7  03                                       1112+          DC    HL1'3'            m4
00001AD8  E5D4D540 40404040                        1113+          DC    CL8'VMN'          instruction name
00001AE0  00001B48                                 1114+          DC    A(RE18+16)        address of v2 source
00001AE4  00001B58                                 1115+          DC    A(RE18+32)        address of v3 source
00001AE8  00000010                                 1116+          DC    A(16)             result length
00001AEC  00001B38                                 1117+REA18     DC    A(RE18)           result address
00001AF0  00000000 00000000                        1118+          DS    FD                gap
00001AF8  00000000 00000000                        1119+V1O18     DS    XL16              V1 output
00001B00  00000000 00000000
00001B08  00000000 00000000                        1120+          DS    FD                gap
                                                   1121+*
00001B10                                           1122+X18       DS    0F
00001B10  E310 5010 0014              00000010     1123+          LGF   R1,V2ADDR         load v2 source
00001B16  E761 0000 0806              00000000     1124+          VL    v22,0(R1)         use v22 to test decoder
00001B1C  E310 5014 0014              00000014     1125+          LGF   R1,V3ADDR         load v3 source
00001B22  E771 0000 0806              00000000     1126+          VL    v23,0(R1)         use v23 to test decoder
00001B28  E766 7000 3EFE                           1127+          VMN   V22,V22,V23,3      test instruction (dest is a source)
00001B2E  E760 5028 080E              00001AF8     1128+          VST   V22,V1O18          save v1 output
00001B34  07FB                                     1129+          BR    R11               return
00001B38                                           1130+RE18      DC    0F                xl16 expected result
00001B38                                           1131+          DROP  R5
00001B38  FFFFFFFF FFFFFFFD                        1132           DC    XL16'FFFFFFFFFFFFFFFD000000000000001F'   expected result
00001B40  00000000 0000001F
00001B48  FFFFFFFF FFFFFFFF                        1133           DC    XL16'FFFFFFFFFFFFFFFF000000000000001F'   v2
00001B50  00000000 0000001F
00001B58  FFFFFFFF FFFFFFFD                        1134           DC    XL16'FFFFFFFFFFFFFFFD0000000000000020'   v3
00001B60  00000000 00000020
                                                   1135
                                                   1136
                                                   1137 * quadword
                                                   1138           VRR_C VMN,4
00001B68                                           1139+          DS    0FD
00001B68                     00001B68              1140+          USING *,R5              base for test data and test routine
00001B68  00001BA8                                 1141+T19       DC    A(X19)            address of test routine
00001B6C  0013                                     1142+          DC    H'19'             test number
00001B6E  00                                       1143+          DC    X'00'
00001B6F  04                                       1144+          DC    HL1'4'            m4
00001B70  E5D4D540 40404040                        1145+          DC    CL8'VMN'          instruction name
00001B78  00001BE0                                 1146+          DC    A(RE19+16)        address of v2 source
00001B7C  00001BF0                                 1147+          DC    A(RE19+32)        address of v3 source
00001B80  00000010                                 1148+          DC    A(16)             result length
00001B84  00001BD0                                 1149+REA19     DC    A(RE19)           result address
00001B88  00000000 00000000                        1150+          DS    FD                gap
00001B90  00000000 00000000                        1151+V1O19     DS    XL16              V1 output
```

```
  LOC       OBJECT CODE       ADDR1     ADDR2     STMT

00001B98  00000000 00000000
00001BA0  00000000 00000000                       1152+          DS    FD                 gap
                                                  1153+*
00001BA8                                          1154+X19       DS    0F
00001BA8  E310 5010 0014               00000010   1155+          LGF   R1,V2ADDR          load v2 source
00001BAE  E761 0000 0806               00000000   1156+          VL    v22,0(R1)          use v22 to test decoder
00001BB4  E310 5014 0014               00000014   1157+          LGF   R1,V3ADDR          load v3 source
00001BBA  E771 0000 0806               00000000   1158+          VL    v23,0(R1)          use v23 to test decoder
00001BC0  E766 7000 4EFE                          1159+          VMN   V22,V22,V23,4       test instruction (dest is a source)
00001BC6  E760 5028 080E               00001B90   1160+          VST   V22,V1O19           save v1 output
00001BCC  07FB                                    1161+          BR    R11                return
00001BD0                                          1162+RE19      DC    0F                 xl16 expected result
00001BD0                                          1163+          DROP  R5
00001BD0  FFFFFFFF FFFFFFFD                       1164           DC    XL16'FFFFFFFFFFFFFFFD0000000000000020'   expected result
00001BD8  00000000 00000020
00001BE0  FFFFFFFF FFFFFFFF                       1165           DC    XL16'FFFFFFFFFFFFFFFF000000000000001F'   v2
00001BE8  00000000 0000001F
00001BF0  FFFFFFFF FFFFFFFD                       1166           DC    XL16'FFFFFFFFFFFFFFFD0000000000000020'   v3
00001BF8  00000000 00000020
                                                  1167
                                                  1168 * quadword
                                                  1169           VRR_C VMN,4
00001C00                                          1170+          DS    0FD
00001C00                        00001C00          1171+          USING *,R5               base for test data and test routine
00001C00  00001C40                                1172+T20       DC    A(X20)             address of test routine
00001C04  0014                                    1173+          DC    H'20'              test number
00001C06  00                                      1174+          DC    X'00'
00001C07  04                                      1175+          DC    HL1'4'             m4
00001C08  E5D4D540 40404040                       1176+          DC    CL8'VMN'           instruction name
00001C10  00001C78                                1177+          DC    A(RE20+16)         address of v2 source
00001C14  00001C88                                1178+          DC    A(RE20+32)         address of v3 source
00001C18  00000010                                1179+          DC    A(16)              result length
00001C1C  00001C68                                1180+REA20     DC    A(RE20)            result address
00001C20  00000000 00000000                       1181+          DS    FD                 gap
00001C28  00000000 00000000                       1182+V1O20     DS    XL16               V1 output
00001C30  00000000 00000000
00001C38  00000000 00000000                       1183+          DS    FD                 gap
                                                  1184+*
00001C40                                          1185+X20       DS    0F
00001C40  E310 5010 0014               00000010   1186+          LGF   R1,V2ADDR          load v2 source
00001C46  E761 0000 0806               00000000   1187+          VL    v22,0(R1)          use v22 to test decoder
00001C4C  E310 5014 0014               00000014   1188+          LGF   R1,V3ADDR          load v3 source
00001C52  E771 0000 0806               00000000   1189+          VL    v23,0(R1)          use v23 to test decoder
00001C58  E766 7000 4EFE                          1190+          VMN   V22,V22,V23,4       test instruction (dest is a source)
00001C5E  E760 5028 080E               00001C28   1191+          VST   V22,V1O20           save v1 output
00001C64  07FB                                    1192+          BR    R11                return
00001C68                                          1193+RE20      DC    0F                 xl16 expected result
00001C68                                          1194+          DROP  R5
00001C68  FFFFFFFF FFFFFFFD                       1195           DC    XL16'FFFFFFFFFFFFFFFD0000000000000020'  expected result
00001C70  00000000 00000020
00001C78  0001FFFF FFFD8000                       1196           DC    XL16'0001FFFFFFFD80007FFF80000123001F'   v2
00001C80  7FFF8000 0123001F
00001C88  FFFFFFFF FFFFFFFD                       1197           DC    XL16'FFFFFFFFFFFFFFFD0000000000000020'   v3
00001C90  00000000 00000020
                                                  1198
                                                  1199 * quadword
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                  1200          VRR_C VMN,4
00001C98                                          1201+         DS    0FD
00001C98                      00001C98            1202+         USING *,R5               base for test data and test routine
00001C98  00001CD8                                1203+T21      DC    A(X21)             address of test routine
00001C9C  0015                                    1204+         DC    H'21'              test number
00001C9E  00                                      1205+         DC    X'00'
00001C9F  04                                      1206+         DC    HL1'4'             m4
00001CA0  E5D4D540 40404040                       1207+         DC    CL8'VMN'           instruction name
00001CA8  00001D10                                1208+         DC    A(RE21+16)         address of v2 source
00001CAC  00001D20                                1209+         DC    A(RE21+32)         address of v3 source
00001CB0  00000010                                1210+         DC    A(16)              result length
00001CB4  00001D00                                1211+REA21    DC    A(RE21)            result address
00001CB8  00000000 00000000                       1212+         DS    FD                 gap
00001CC0  00000000 00000000                       1213+V1O21    DS    XL16               V1 output
00001CC8  00000000 00000000
00001CD0  00000000 00000000                       1214+         DS    FD                 gap
                                                  1215+*
00001CD8                                          1216+X21      DS    0F
00001CD8  E310 5010 0014             00000010     1217+         LGF   R1,V2ADDR          load v2 source
00001CDE  E761 0000 0806             00000000     1218+         VL    v22,0(R1)          use v22 to test decoder
00001CE4  E310 5014 0014             00000014     1219+         LGF   R1,V3ADDR          load v3 source
00001CEA  E771 0000 0806             00000000     1220+         VL    v23,0(R1)          use v23 to test decoder
00001CF0  E766 7000 4EFE                          1221+         VMN   V22,V22,V23,4       test instruction (dest is a source)
00001CF6  E760 5028 080E             00001CC0     1222+         VST   V22,V1O21           save v1 output
00001CFC  07FB                                    1223+         BR    R11                return
00001D00                                          1224+RE21     DC    0F                 xl16 expected result
00001D00                                          1225+         DROP  R5
00001D00  0001FFFF FFFD8000                       1226          DC    XL16'0001FFFFFFFD80007FFF80000123001F'   expected result
00001D08  7FFF8000 0123001F
00001D10  0001FFFF FFFD8000                       1227          DC    XL16'0001FFFFFFFD80007FFF80000123001F'   v2
00001D18  7FFF8000 0123001F
00001D20  00020001 FFFE0001                       1228          DC    XL16'00020001FFFE000100AA800112340020'   v3
00001D28  00AA8001 12340020
                                                  1229
                                                  1230 *----------------------------------------------------------------------
                                                  1231 * VMNL   - VECTOR MINIMUM LOGICAL
                                                  1232 *----------------------------------------------------------------------
                                                  1233 * Byte
                                                  1234          VRR_C VMNL,0
00001D30                                          1235+         DS    0FD
00001D30                      00001D30            1236+         USING *,R5               base for test data and test routine
00001D30  00001D70                                1237+T22      DC    A(X22)             address of test routine
00001D34  0016                                    1238+         DC    H'22'              test number
00001D36  00                                      1239+         DC    X'00'
00001D37  00                                      1240+         DC    HL1'0'             m4
00001D38  E5D4D5D3 40404040                       1241+         DC    CL8'VMNL'          instruction name
00001D40  00001DA8                                1242+         DC    A(RE22+16)         address of v2 source
00001D44  00001DB8                                1243+         DC    A(RE22+32)         address of v3 source
00001D48  00000010                                1244+         DC    A(16)              result length
00001D4C  00001D98                                1245+REA22    DC    A(RE22)            result address
00001D50  00000000 00000000                       1246+         DS    FD                 gap
00001D58  00000000 00000000                       1247+V1O22    DS    XL16               V1 output
00001D60  00000000 00000000
00001D68  00000000 00000000                       1248+         DS    FD                 gap
                                                  1249+*
00001D70                                          1250+X22      DS    0F
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00001D70  E310 5010 0014                00000010  1251+          LGF   R1,V2ADDR          load v2 source
00001D76  E761 0000 0806                00000000  1252+          VL    v22,0(R1)          use v22 to test decoder
00001D7C  E310 5014 0014                00000014  1253+          LGF   R1,V3ADDR          load v3 source
00001D82  E771 0000 0806                00000000  1254+          VL    v23,0(R1)          use v23 to test decoder
00001D88  E766 7000 0EFC                          1255+          VMNL  V22,V22,V23,0       test instruction (dest is a source)
00001D8E  E760 5028 080E                00001D58  1256+          VST   V22,V1O22           save v1 output
00001D94  07FB                                    1257+          BR    R11                return
00001D98                                          1258+RE22      DC    0F                 xl16 expected result
00001D98                                          1259+          DROP  R5
00001D98  01020304 00010180                       1260           DC    XL16'0102030400010180010AFEFD0000001F'   expected result
00001DA0  010AFEFD 0000001F
00001DA8  01020304 09800181                       1261           DC    XL16'010203040980018107OFFFFD0000001F'   v2
00001DB0  070FFFFD 0000001F
00001DB8  02030405 0001FF80                       1262           DC    XL16'020304050001FF80010AFEFE00000020'   v3
00001DC0  010AFEFE 00000020
                                                  1263
                                                  1264 * Halfword
                                                  1265           VRR_C VMNL,1
00001DC8                                          1266+          DS    0FD
00001DC8                     00001DC8             1267+          USING *,R5               base for test data and test routine
00001DC8  00001E08                                1268+T23       DC    A(X23)             address of test routine
00001DCC  0017                                    1269+          DC    H'23'              test number
00001DCE  00                                      1270+          DC    X'00'
00001DCF  01                                      1271+          DC    HL1'1'             m4
00001DD0  E5D4D5D3 40404040                       1272+          DC    CL8'VMNL'          instruction name
00001DD8  00001E40                                1273+          DC    A(RE23+16)         address of v2 source
00001DDC  00001E50                                1274+          DC    A(RE23+32)         address of v3 source
00001DE0  00000010                                1275+          DC    A(16)              result length
00001DE4  00001E30                                1276+REA23     DC    A(RE23)            result address
00001DE8  00000000 00000000                       1277+          DS    FD                 gap
00001DF0  00000000 00000000                       1278+V1O23     DS    XL16               V1 output
00001DF8  00000000 00000000
00001E00  00000000 00000000                       1279+          DS    FD                 gap
                                                  1280+*
00001E08                                          1281+X23       DS    0F
00001E08  E310 5010 0014                00000010  1282+          LGF   R1,V2ADDR          load v2 source
00001E0E  E761 0000 0806                00000000  1283+          VL    v22,0(R1)          use v22 to test decoder
00001E14  E310 5014 0014                00000014  1284+          LGF   R1,V3ADDR          load v3 source
00001E1A  E771 0000 0806                00000000  1285+          VL    v23,0(R1)          use v23 to test decoder
00001E20  E766 7000 1EFC                          1286+          VMNL  V22,V22,V23,1       test instruction (dest is a source)
00001E26  E760 5028 080E                00001DF0  1287+          VST   V22,V1O23           save v1 output
00001E2C  07FB                                    1288+          BR    R11                return
00001E30                                          1289+RE23      DC    0F                 xl16 expected result
00001E30                                          1290+          DROP  R5
00001E30  00010001 FFFD0001                       1291           DC    XL16'00010001FFFD000100AA80000123001F'   expected result
00001E38  00AA8000 0123001F
00001E40  0001FFFF FFFD8000                       1292           DC    XL16'0001FFFFFFFD80007FFF80000123001F'   v2
00001E48  7FFF8000 0123001F
00001E50  00020001 FFFE0001                       1293           DC    XL16'00020001FFFE000100AA800112340020'   v3
00001E58  00AA8001 12340020
                                                  1294
                                                  1295 * Word
                                                  1296           VRR_C VMNL,2
00001E60                                          1297+          DS    0FD
00001E60                     00001E60             1298+          USING *,R5               base for test data and test routine
00001E60  00001EA0                                1299+T24       DC    A(X24)             address of test routine
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00001E64  0018                                     1300+          DC    H'24'               test number
00001E66  00                                       1301+          DC    X'00'
00001E67  02                                       1302+          DC    HL1'2'              m4
00001E68  E5D4D5D3 40404040                        1303+          DC    CL8'VMNL'           instruction name
00001E70  00001ED8                                 1304+          DC    A(RE24+16)          address of v2 source
00001E74  00001EE8                                 1305+          DC    A(RE24+32)          address of v3 source
00001E78  00000010                                 1306+          DC    A(16)               result length
00001E7C  00001EC8                                 1307+REA24     DC    A(RE24)             result address
00001E80  00000000 00000000                        1308+          DS    FD                  gap
00001E88  00000000 00000000                        1309+V1O24     DS    XL16                V1 output
00001E90  00000000 00000000
00001E98  00000000 00000000                        1310+          DS    FD                  gap
                                                   1311+*
00001EA0                                           1312+X24       DS    0F
00001EA0  E310 5010 0014              00000010     1313+          LGF   R1,V2ADDR           load v2 source
00001EA6  E761 0000 0806              00000000     1314+          VL    v22,0(R1)           use v22 to test decoder
00001EAC  E310 5014 0014              00000014     1315+          LGF   R1,V3ADDR           load v3 source
00001EB2  E771 0000 0806              00000000     1316+          VL    v23,0(R1)           use v23 to test decoder
00001EB8  E766 7000 2EFC                           1317+          VMNL  V22,V22,V23,2        test instruction (dest is a source)
00001EBE  E760 5028 080E              00001E88     1318+          VST   V22,V1O24            save v1 output
00001EC4  07FB                                     1319+          BR    R11                 return
00001EC8                                           1320+RE24      DC    0F                  xl16 expected result
00001EC8                                           1321+          DROP  R5
00001EC8  FFFFFFFE 0000000A                        1322           DC    XL16'FFFFFFFE0000000A012345670000001F'   expected result
00001ED0  01234567 0000001F
00001ED8  FFFFFFFF 7FFFFFFF                        1323           DC    XL16'FFFFFFFF7FFFFFFF012345670000001F'   v2
00001EE0  01234567 0000001F
00001EE8  FFFFFFFE 0000000A                        1324           DC    XL16'FFFFFFFE0000000A1234567800000020'   v3
00001EF0  12345678 00000020
                                                   1325
                                                   1326 * Doubleword
                                                   1327           VRR_C VMNL,3
00001EF8                                           1328+          DS    0FD
00001EF8                     00001EF8              1329+          USING *,R5              base for test data and test routine
00001EF8  00001F38                                 1330+T25       DC    A(X25)              address of test routine
00001EFC  0019                                     1331+          DC    H'25'               test number
00001EFE  00                                       1332+          DC    X'00'
00001EFF  03                                       1333+          DC    HL1'3'              m4
00001F00  E5D4D5D3 40404040                        1334+          DC    CL8'VMNL'           instruction name
00001F08  00001F70                                 1335+          DC    A(RE25+16)          address of v2 source
00001F0C  00001F80                                 1336+          DC    A(RE25+32)          address of v3 source
00001F10  00000010                                 1337+          DC    A(16)               result length
00001F14  00001F60                                 1338+REA25     DC    A(RE25)             result address
00001F18  00000000 00000000                        1339+          DS    FD                  gap
00001F20  00000000 00000000                        1340+V1O25     DS    XL16                V1 output
00001F28  00000000 00000000
00001F30  00000000 00000000                        1341+          DS    FD                  gap
                                                   1342+*
00001F38                                           1343+X25       DS    0F
00001F38  E310 5010 0014              00000010     1344+          LGF   R1,V2ADDR           load v2 source
00001F3E  E761 0000 0806              00000000     1345+          VL    v22,0(R1)           use v22 to test decoder
00001F44  E310 5014 0014              00000014     1346+          LGF   R1,V3ADDR           load v3 source
00001F4A  E771 0000 0806              00000000     1347+          VL    v23,0(R1)           use v23 to test decoder
00001F50  E766 7000 3EFC                           1348+          VMNL  V22,V22,V23,3        test instruction (dest is a source)
00001F56  E760 5028 080E              00001F20     1349+          VST   V22,V1O25            save v1 output
00001F5C  07FB                                     1350+          BR    R11                 return
```

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  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00001F60                                           1351+RE25     DC    0F                 xl16 expected result
00001F60                                           1352+         DROP  R5
00001F60  FFFFFFFF FFFFFFFD                        1353          DC    XL16'FFFFFFFFFFFFFFFD000000000000001F'   expected result
00001F68  00000000 0000001F
00001F70  FFFFFFFF FFFFFFFF                        1354          DC    XL16'FFFFFFFFFFFFFFFF000000000000001F'   v2
00001F78  00000000 0000001F
00001F80  FFFFFFFF FFFFFFFD                        1355          DC    XL16'FFFFFFFFFFFFFFFD0000000000000020'   v3
00001F88  00000000 00000020
                                                   1356
                                                   1357 * quadword
                                                   1358          VRR_C VMNL,4
00001F90                                           1359+         DS    0FD
00001F90                     00001F90              1360+         USING *,R5               base for test data and test routine
00001F90  00001FD0                                 1361+T26      DC    A(X26)             address of test routine
00001F94  001A                                     1362+         DC    H'26'              test number
00001F96  00                                       1363+         DC    X'00'
00001F97  04                                       1364+         DC    HL1'4'             m4
00001F98  E5D4D5D3 40404040                        1365+         DC    CL8'VMNL'          instruction name
00001FA0  00002008                                 1366+         DC    A(RE26+16)         address of v2 source
00001FA4  00002018                                 1367+         DC    A(RE26+32)         address of v3 source
00001FA8  00000010                                 1368+         DC    A(16)              result length
00001FAC  00001FF8                                 1369+REA26    DC    A(RE26)            result address
00001FB0  00000000 00000000                        1370+         DS    FD                 gap
00001FB8  00000000 00000000                        1371+V1O26    DS    XL16               V1 output
00001FC0  00000000 00000000
00001FC8  00000000 00000000                        1372+         DS    FD                 gap
                                                   1373+*
00001FD0                                           1374+X26      DS    0F
00001FD0  E310 5010 0014               00000010    1375+         LGF   R1,V2ADDR          load v2 source
00001FD6  E761 0000 0806               00000000    1376+         VL    v22,0(R1)          use v22 to test decoder
00001FDC  E310 5014 0014               00000014    1377+         LGF   R1,V3ADDR          load v3 source
00001FE2  E771 0000 0806               00000000    1378+         VL    v23,0(R1)          use v23 to test decoder
00001FE8  E766 7000 4EFC                           1379+         VMNL  V22,V22,V23,4       test instruction (dest is a source)
00001FEE  E760 5028 080E               00001FB8    1380+         VST   V22,V1O26           save v1 output
00001FF4  07FB                                     1381+         BR    R11                return
00001FF8                                           1382+RE26     DC    0F                 xl16 expected result
00001FF8                                           1383+         DROP  R5
00001FF8  FFFFFFFF FFFFFFFD                        1384          DC    XL16'FFFFFFFFFFFFFFFD0000000000000020'   expected result
00002000  00000000 00000020
00002008  FFFFFFFF FFFFFFFF                        1385          DC    XL16'FFFFFFFFFFFFFFFF000000000000001F'   v2
00002010  00000000 0000001F
00002018  FFFFFFFF FFFFFFFD                        1386          DC    XL16'FFFFFFFFFFFFFFFD0000000000000020'   v3
00002020  00000000 00000020
                                                   1387
                                                   1388 * quadword
                                                   1389          VRR_C VMNL,4
00002028                                           1390+         DS    0FD
00002028                     00002028              1391+         USING *,R5               base for test data and test routine
00002028  00002068                                 1392+T27      DC    A(X27)             address of test routine
0000202C  001B                                     1393+         DC    H'27'              test number
0000202E  00                                       1394+         DC    X'00'
0000202F  04                                       1395+         DC    HL1'4'             m4
00002030  E5D4D5D3 40404040                        1396+         DC    CL8'VMNL'          instruction name
00002038  000020A0                                 1397+         DC    A(RE27+16)         address of v2 source
0000203C  000020B0                                 1398+         DC    A(RE27+32)         address of v3 source
00002040  00000010                                 1399+         DC    A(16)              result length
```

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  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002044  00002090                                 1400+REA27    DC    A(RE27)             result address
00002048  00000000 00000000                        1401+         DS    FD                  gap
00002050  00000000 00000000                        1402+V1O27    DS    XL16                V1 output
00002058  00000000 00000000
00002060  00000000 00000000                        1403+         DS    FD                  gap
                                                   1404+*
00002068                                           1405+X27      DS    0F
00002068  E310 5010 0014             00000010      1406+         LGF   R1,V2ADDR           load v2 source
0000206E  E761 0000 0806             00000000      1407+         VL    v22,0(R1)           use v22 to test decoder
00002074  E310 5014 0014             00000014      1408+         LGF   R1,V3ADDR           load v3 source
0000207A  E771 0000 0806             00000000      1409+         VL    v23,0(R1)           use v23 to test decoder
00002080  E766 7000 4EFC                           1410+         VMNL  V22,V22,V23,4        test instruction (dest is a source)
00002086  E760 5028 080E             00002050      1411+         VST   V22,V1O27            save v1 output
0000208C  07FB                                     1412+         BR    R11                 return
00002090                                           1413+RE27     DC    0F                  xl16 expected result
00002090                                           1414+         DROP  R5
00002090  0001FFFF FFFD8000                        1415          DC    XL16'0001FFFFFFFD80007FFF80000123001F'  expected result
00002098  7FFF8000 0123001F
000020A0  0001FFFF FFFD8000                        1416          DC    XL16'0001FFFFFFFD80007FFF80000123001F'   v2
000020A8  7FFF8000 0123001F
000020B0  FFFFFFFF FFFFFFFD                        1417          DC    XL16'FFFFFFFFFFFFFFFD0000000000000020'   v3
000020B8  00000000 00000020
                                                   1418
                                                   1419 * quadword
                                                   1420          VRR_C VMNL,4
000020C0                                           1421+         DS    0FD
000020C0                       000020C0            1422+         USING *,R5                base for test data and test routine
000020C0  00002100                                 1423+T28      DC    A(X28)              address of test routine
000020C4  001C                                     1424+         DC    H'28'               test number
000020C6  00                                       1425+         DC    X'00'
000020C7  04                                       1426+         DC    HL1'4'              m4
000020C8  E5D4D5D3 40404040                        1427+         DC    CL8'VMNL'           instruction name
000020D0  00002138                                 1428+         DC    A(RE28+16)          address of v2 source
000020D4  00002148                                 1429+         DC    A(RE28+32)          address of v3 source
000020D8  00000010                                 1430+         DC    A(16)               result length
000020DC  00002128                                 1431+REA28    DC    A(RE28)             result address
000020E0  00000000 00000000                        1432+         DS    FD                  gap
000020E8  00000000 00000000                        1433+V1O28    DS    XL16                V1 output
000020F0  00000000 00000000
000020F8  00000000 00000000                        1434+         DS    FD                  gap
                                                   1435+*
00002100                                           1436+X28      DS    0F
00002100  E310 5010 0014             00000010      1437+         LGF   R1,V2ADDR           load v2 source
00002106  E761 0000 0806             00000000      1438+         VL    v22,0(R1)           use v22 to test decoder
0000210C  E310 5014 0014             00000014      1439+         LGF   R1,V3ADDR           load v3 source
00002112  E771 0000 0806             00000000      1440+         VL    v23,0(R1)           use v23 to test decoder
00002118  E766 7000 4EFC                           1441+         VMNL  V22,V22,V23,4        test instruction (dest is a source)
0000211E  E760 5028 080E             000020E8      1442+         VST   V22,V1O28            save v1 output
00002124  07FB                                     1443+         BR    R11                 return
00002128                                           1444+RE28     DC    0F                  xl16 expected result
00002128                                           1445+         DROP  R5
00002128  0001FFFF FFFD8000                        1446          DC    XL16'0001FFFFFFFD80007FFF80000123001F'   expected result
00002130  7FFF8000 0123001F
00002138  0001FFFF FFFD8000                        1447          DC    XL16'0001FFFFFFFD80007FFF80000123001F'   v2
00002140  7FFF8000 0123001F
00002148  00020001 FFFE0001                        1448          DC    XL16'00020001FFFE000100AA800112340020'   v3
```

```
  LOC       OBJECT CODE       ADDR1     ADDR2     STMT

00002150  00AA8001 12340020
                                                  1449
                                                  1450 *--------------------------------------------------------------------
                                                  1451 * VAVG    - VECTOR AVERAGE
                                                  1452 *--------------------------------------------------------------------
                                                  1453 * Byte
                                                  1454          VRR_C VAVG,0
00002158                                          1455+         DS    0FD
00002158                            00002158      1456+         USING *,R5               base for test data and test routine
00002158  00002198                                1457+T29      DC    A(X29)             address of test routine
0000215C  001D                                    1458+         DC    H'29'              test number
0000215E  00                                      1459+         DC    X'00'
0000215F  00                                      1460+         DC    HL1'0'             m4
00002160  E5C1E5C7 40404040                       1461+         DC    CL8'VAVG'          instruction name
00002168  000021D0                                1462+         DC    A(RE29+16)         address of v2 source
0000216C  000021E0                                1463+         DC    A(RE29+32)         address of v3 source
00002170  00000010                                1464+         DC    A(16)              result length
00002174  000021C0                                1465+REA29    DC    A(RE29)            result address
00002178  00000000 00000000                       1466+         DS    FD                 gap
00002180  00000000 00000000                       1467+V1O29    DS    XL16               V1 output
00002188  00000000 00000000
00002190  00000000 00000000                       1468+         DS    FD                 gap
                                                  1469+*
00002198                                          1470+X29      DS    0F
00002198  E310 5010 0014              00000010    1471+         LGF   R1,V2ADDR          load v2 source
0000219E  E761 0000 0806              00000000    1472+         VL    v22,0(R1)          use v22 to test decoder
000021A4  E310 5014 0014              00000014    1473+         LGF   R1,V3ADDR          load v3 source
000021AA  E771 0000 0806              00000000    1474+         VL    v23,0(R1)          use v23 to test decoder
000021B0  E766 7000 0EF2                          1475+         VAVG  V22,V22,V23,0       test instruction (dest is a source)
000021B6  E760 5028 080E              00002180    1476+         VST   V22,V1O29           save v1 output
000021BC  07FB                                    1477+         BR    R11                return
000021C0                                          1478+RE29     DC    0F                 xl16 expected result
000021C0                                          1479+         DROP  R5
000021C0  FFFFFFFF FFFFFFFF                       1480          DC    XL16'FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF'   expected result
000021C8  FFFFFFFF FFFFFFFF
000021D0  7C7C7C7C 7C7C7C7C                       1481          DC    XL16'7C7C7C7C7C7C7C7C7C7C7C7C7C7C7C7C'   v2
000021D8  7C7C7C7C 7C7C7C7C
000021E0  82828282 82828282                       1482          DC    XL16'82828282828282828282828282828282'   v3
000021E8  82828282 82828282
                                                  1483
                                                  1484 * Halfword
                                                  1485          VRR_C VAVG,1
000021F0                                          1486+         DS    0FD
000021F0                            000021F0      1487+         USING *,R5               base for test data and test routine
000021F0  00002230                                1488+T30      DC    A(X30)             address of test routine
000021F4  001E                                    1489+         DC    H'30'              test number
000021F6  00                                      1490+         DC    X'00'
000021F7  01                                      1491+         DC    HL1'1'             m4
000021F8  E5C1E5C7 40404040                       1492+         DC    CL8'VAVG'          instruction name
00002200  00002268                                1493+         DC    A(RE30+16)         address of v2 source
00002204  00002278                                1494+         DC    A(RE30+32)         address of v3 source
00002208  00000010                                1495+         DC    A(16)              result length
0000220C  00002258                                1496+REA30    DC    A(RE30)            result address
00002210  00000000 00000000                       1497+         DS    FD                 gap
00002218  00000000 00000000                       1498+V1O30    DS    XL16               V1 output
00002220  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002228  00000000 00000000                        1499+         DS    FD                 gap
                                                   1500+*
00002230                                           1501+X30      DS    0F
00002230  E310 5010 0014                00000010  1502+         LGF   R1,V2ADDR          load v2 source
00002236  E761 0000 0806                00000000  1503+         VL    v22,0(R1)          use v22 to test decoder
0000223C  E310 5014 0014                00000014  1504+         LGF   R1,V3ADDR          load v3 source
00002242  E771 0000 0806                00000000  1505+         VL    v23,0(R1)          use v23 to test decoder
00002248  E766 7000 1EF2                          1506+         VAVG  V22,V22,V23,1       test instruction (dest is a source)
0000224E  E760 A018 080E                00002218  1507+         VST   V22,V1O30           save v1 output
00002254  07FB                                    1508+         BR    R11                return
00002258                                          1509+RE30     DC    0F                 xl16 expected result
00002258                                          1510+         DROP  R5
00002258  FF7FFF7F FF7FFF7F                       1511          DC    XL16'FF7FFF7FFF7FFF7FFF7FFF7FFF7FFF7F'   expected result
00002260  FF7FFF7F FF7FFF7F
00002268  7C7C7C7C 7C7C7C7C                       1512          DC    XL16'7C7C7C7C7C7C7C7C7C7C7C7C7C7C7C7C'   v2
00002270  7C7C7C7C 7C7C7C7C
00002278  82828282 82828282                       1513          DC    XL16'82828282828282828282828282828282'   v3
00002280  82828282 82828282
                                                  1514
                                                  1515 * Word
                                                  1516          VRR_C VAVG,2
00002288                                          1517+         DS    0FD
00002288                     00002288             1518+         USING *,R5               base for test data and test routine
00002288  000022C8                                1519+T31      DC    A(X31)             address of test routine
0000228C  001F                                    1520+         DC    H'31'              test number
0000228E  00                                      1521+         DC    X'00'
0000228F  02                                      1522+         DC    HL1'2'             m4
00002290  E5C1E5C7 40404040                       1523+         DC    CL8'VAVG'          instruction name
00002298  00002300                                1524+         DC    A(RE31+16)         address of v2 source
0000229C  00002310                                1525+         DC    A(RE31+32)         address of v3 source
000022A0  00000010                                1526+         DC    A(16)              result length
000022A4  000022F0                                1527+REA31    DC    A(RE31)            result address
000022A8  00000000 00000000                       1528+         DS    FD                 gap
000022B0  00000000 00000000                       1529+V1O31    DS    XL16               V1 output
000022B8  00000000 00000000
000022C0  00000000 00000000                       1530+         DS    FD                 gap
                                                  1531+*
000022C8                                          1532+X31      DS    0F
000022C8  E310 5010 0014                00000010  1533+         LGF   R1,V2ADDR          load v2 source
000022CE  E761 0000 0806                00000000  1534+         VL    v22,0(R1)          use v22 to test decoder
000022D4  E310 5014 0014                00000014  1535+         LGF   R1,V3ADDR          load v3 source
000022DA  E771 0000 0806                00000000  1536+         VL    v23,0(R1)          use v23 to test decoder
000022E0  E766 7000 2EF2                          1537+         VAVG  V22,V22,V23,2       test instruction (dest is a source)
000022E6  E760 5028 080E                000022B0  1538+         VST   V22,V1O31           save v1 output
000022EC  07FB                                    1539+         BR    R11                return
000022F0                                          1540+RE31     DC    0F                 xl16 expected result
000022F0                                          1541+         DROP  R5
000022F0  FF7F7F7F FF7F7F7F                       1542          DC    XL16'FF7F7F7FFF7F7F7FFF7F7F7FFF7F7F7F'   expected result
000022F8  FF7F7F7F FF7F7F7F
00002300  7C7C7C7C 7C7C7C7C                       1543          DC    XL16'7C7C7C7C7C7C7C7C7C7C7C7C7C7C7C7C'   v2
00002308  7C7C7C7C 7C7C7C7C
00002310  82828282 82828282                       1544          DC    XL16'82828282828282828282828282828282'   v3
00002318  82828282 82828282
                                                  1545
                                                  1546 * Doubleword
                                                  1547          VRR_C VAVG,3
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002320                                           1548+          DS    0FD
00002320                       00002320            1549+          USING *,R5                base for test data and test routine
00002320  00002360                                 1550+T32       DC    A(X32)              address of test routine
00002324  0020                                     1551+          DC    H'32'               test number
00002326  00                                       1552+          DC    X'00'
00002327  03                                       1553+          DC    HL1'3'              m4
00002328  E5C1E5C7 40404040                        1554+          DC    CL8'VAVG'           instruction name
00002330  00002398                                 1555+          DC    A(RE32+16)          address of v2 source
00002334  000023A8                                 1556+          DC    A(RE32+32)          address of v3 source
00002338  00000010                                 1557+          DC    A(16)               result length
0000233C  00002388                                 1558+REA32     DC    A(RE32)             result address
00002340  00000000 00000000                        1559+          DS    FD                  gap
00002348  00000000 00000000                        1560+V1O32     DS    XL16                V1 output
00002350  00000000 00000000
00002358  00000000 00000000                        1561+          DS    FD                  gap
                                                   1562+*
00002360                                           1563+X32       DS    0F
00002360  E310 5010 0014              00000010     1564+          LGF   R1,V2ADDR           load v2 source
00002366  E761 0000 0806              00000000     1565+          VL    v22,0(R1)           use v22 to test decoder
0000236C  E310 5014 0014              00000014     1566+          LGF   R1,V3ADDR           load v3 source
00002372  E771 0000 0806              00000000     1567+          VL    v23,0(R1)           use v23 to test decoder
00002378  E766 7000 3EF2                           1568+          VAVG  V22,V22,V23,3        test instruction (dest is a source)
0000237E  E760 5028 080E              00002348     1569+          VST   V22,V1O32            save v1 output
00002384  07FB                                     1570+          BR    R11                 return
00002388                                           1571+RE32      DC    0F                  xl16 expected result
00002388                                           1572+          DROP  R5
00002388  FF7F7F7F 7F7F7F7F                        1573           DC    XL16'FF7F7F7F7F7F7F7FFF7F7F7F7F7F7F7F'   expected result
00002390  FF7F7F7F 7F7F7F7F
00002398  7C7C7C7C 7C7C7C7C                        1574           DC    XL16'7C7C7C7C7C7C7C7C7C7C7C7C7C7C7C7C'   v2
000023A0  7C7C7C7C 7C7C7C7C
000023A8  82828282 82828282                        1575           DC    XL16'82828282828282828282828282828282'   v3
000023B0  82828282 82828282
                                                   1576
                                                   1577 * Doubleword
                                                   1578           VRR_C VAVG,3
000023B8                                           1579+          DS    0FD
000023B8                       000023B8            1580+          USING *,R5                base for test data and test routine
000023B8  000023F8                                 1581+T33       DC    A(X33)              address of test routine
000023BC  0021                                     1582+          DC    H'33'               test number
000023BE  00                                       1583+          DC    X'00'
000023BF  03                                       1584+          DC    HL1'3'              m4
000023C0  E5C1E5C7 40404040                        1585+          DC    CL8'VAVG'           instruction name
000023C8  00002430                                 1586+          DC    A(RE33+16)          address of v2 source
000023CC  00002440                                 1587+          DC    A(RE33+32)          address of v3 source
000023D0  00000010                                 1588+          DC    A(16)               result length
000023D4  00002420                                 1589+REA33     DC    A(RE33)             result address
000023D8  00000000 00000000                        1590+          DS    FD                  gap
000023E0  00000000 00000000                        1591+V1O33     DS    XL16                V1 output
000023E8  00000000 00000000
000023F0  00000000 00000000                        1592+          DS    FD                  gap
                                                   1593+*
000023F8                                           1594+X33       DS    0F
000023F8  E310 5010 0014              00000010     1595+          LGF   R1,V2ADDR           load v2 source
000023FE  E761 0000 0806              00000000     1596+          VL    v22,0(R1)           use v22 to test decoder
00002404  E310 5014 0014              00000014     1597+          LGF   R1,V3ADDR           load v3 source
0000240A  E771 0000 0806              00000000     1598+          VL    v23,0(R1)           use v23 to test decoder
```

```
  LOC       OBJECT CODE       ADDR1     ADDR2     STMT

00002410  E766 7000 3EF2                          1599+          VAVG  V22,V22,V23,3      test instruction (dest is a source)
00002416  E760 5028 080E                000023E0  1600+          VST   V22,V1O33          save v1 output
0000241C  07FB                                    1601+          BR    R11               return
00002420                                          1602+RE33      DC    0F                xl16 expected result
00002420                                          1603+          DROP  R5
00002420  7FFFFFFF FFFFFFFC                       1604           DC    XL16'7FFFFFFFFFFFFFFC7FFFFFFFFFFFFFFC'   expected result
00002428  7FFFFFFF FFFFFFFC
00002430  7FFFFFFF FFFFFFFC                       1605           DC    XL16'7FFFFFFFFFFFFFFC7FFFFFFFFFFFFFFC'   v2
00002438  7FFFFFFF FFFFFFFC
00002440  7FFFFFFF FFFFFFFC                       1606           DC    XL16'7FFFFFFFFFFFFFFC7FFFFFFFFFFFFFFC'   v3
00002448  7FFFFFFF FFFFFFFC
                                                  1607
                                                  1608 * Doubleword
                                                  1609           VRR_C VAVG,3
00002450                                          1610+          DS    0FD
00002450                      00002450            1611+          USING *,R5              base for test data and test routine
00002450  00002490                                1612+T34       DC    A(X34)            address of test routine
00002454  0022                                    1613+          DC    H'34'             test number
00002456  00                                      1614+          DC    X'00'
00002457  03                                      1615+          DC    HL1'3'            m4
00002458  E5C1E5C7 40404040                       1616+          DC    CL8'VAVG'         instruction name
00002460  000024C8                                1617+          DC    A(RE34+16)        address of v2 source
00002464  000024D8                                1618+          DC    A(RE34+32)        address of v3 source
00002468  00000010                                1619+          DC    A(16)             result length
0000246C  000024B8                                1620+REA34     DC    A(RE34)           result address
00002470  00000000 00000000                       1621+          DS    FD                gap
00002478  00000000 00000000                       1622+V1O34     DS    XL16              V1 output
00002480  00000000 00000000
00002488  00000000 00000000                       1623+          DS    FD                gap
                                                  1624+*
00002490                                          1625+X34       DS    0F
00002490  E310 5010 0014                00000010  1626+          LGF   R1,V2ADDR         load v2 source
00002496  E761 0000 0806                00000000  1627+          VL    v22,0(R1)         use v22 to test decoder
0000249C  E310 5014 0014                00000014  1628+          LGF   R1,V3ADDR         load v3 source
000024A2  E771 0000 0806                00000000  1629+          VL    v23,0(R1)         use v23 to test decoder
000024A8  E766 7000 3EF2                          1630+          VAVG  V22,V22,V23,3      test instruction (dest is a source)
000024AE  E760 5028 080E                00002478  1631+          VST   V22,V1O34          save v1 output
000024B4  07FB                                    1632+          BR    R11               return
000024B8                                          1633+RE34      DC    0F                xl16 expected result
000024B8                                          1634+          DROP  R5
000024B8  FFFFFFFF FFFFFFFF                       1635           DC    XL16'FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF'   expected result
000024C0  FFFFFFFF FFFFFFFF
000024C8  80000000 00000002                       1636           DC    XL16'80000000000000028000000000000002'   v2
000024D0  80000000 00000002
000024D8  7FFFFFFF FFFFFFFC                       1637           DC    XL16'7FFFFFFFFFFFFFFC7FFFFFFFFFFFFFFC'   v3
000024E0  7FFFFFFF FFFFFFFC
                                                  1638
                                                  1639 * Doubleword
                                                  1640           VRR_C VAVG,3
000024E8                                          1641+          DS    0FD
000024E8                      000024E8            1642+          USING *,R5              base for test data and test routine
000024E8  00002528                                1643+T35       DC    A(X35)            address of test routine
000024EC  0023                                    1644+          DC    H'35'             test number
000024EE  00                                      1645+          DC    X'00'
000024EF  03                                      1646+          DC    HL1'3'            m4
000024F0  E5C1E5C7 40404040                       1647+          DC    CL8'VAVG'         instruction name
```

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  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

000024F8  00002560                                1648+         DC    A(RE35+16)         address of v2 source
000024FC  00002570                                1649+         DC    A(RE35+32)         address of v3 source
00002500  00000010                                1650+         DC    A(16)              result length
00002504  00002550                                1651+REA35    DC    A(RE35)            result address
00002508  00000000 00000000                       1652+         DS    FD                 gap
00002510  00000000 00000000                       1653+V1O35    DS    XL16               V1 output
00002518  00000000 00000000
00002520  00000000 00000000                       1654+         DS    FD                 gap
                                                  1655+*
00002528                                          1656+X35      DS    0F
00002528  E310 5010 0014             00000010     1657+         LGF   R1,V2ADDR          load v2 source
0000252E  E761 0000 0806             00000000     1658+         VL    v22,0(R1)          use v22 to test decoder
00002534  E310 5014 0014             00000014     1659+         LGF   R1,V3ADDR          load v3 source
0000253A  E771 0000 0806             00000000     1660+         VL    v23,0(R1)          use v23 to test decoder
00002540  E766 7000 3EF2                          1661+         VAVG  V22,V22,V23,3       test instruction (dest is a source)
00002546  E760 5028 080E             00002510     1662+         VST   V22,V1O35           save v1 output
0000254C  07FB                                    1663+         BR    R11                return
00002550                                          1664+RE35     DC    0F                 xl16 expected result
00002550                                          1665+         DROP  R5
00002550  FFFFFFFF FFFFFFFF                       1666          DC    XL16'FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF'   expected result
00002558  FFFFFFFF FFFFFFFF
00002560  7FFFFFFF FFFFFFFC                       1667          DC    XL16'7FFFFFFFFFFFFFFC7FFFFFFFFFFFFFFC'    v2
00002568  7FFFFFFF FFFFFFFC
00002570  80000000 00000002                       1668          DC    XL16'80000000000000028000000000000002'   v3
00002578  80000000 00000002
                                                  1669
                                                  1670 * Doubleword
                                                  1671          VRR_C VAVG,3
00002580                                          1672+         DS    0FD
00002580                      00002580            1673+         USING *,R5               base for test data and test routine
00002580  000025C0                                1674+T36      DC    A(X36)             address of test routine
00002584  0024                                    1675+         DC    H'36'              test number
00002586  00                                      1676+         DC    X'00'
00002587  03                                      1677+         DC    HL1'3'             m4
00002588  E5C1E5C7 40404040                       1678+         DC    CL8'VAVG'          instruction name
00002590  000025F8                                1679+         DC    A(RE36+16)         address of v2 source
00002594  00002608                                1680+         DC    A(RE36+32)         address of v3 source
00002598  00000010                                1681+         DC    A(16)              result length
0000259C  000025E8                                1682+REA36    DC    A(RE36)            result address
000025A0  00000000 00000000                       1683+         DS    FD                 gap
000025A8  00000000 00000000                       1684+V1O36    DS    XL16               V1 output
000025B0  00000000 00000000
000025B8  00000000 00000000                       1685+         DS    FD                 gap
                                                  1686+*
000025C0                                          1687+X36      DS    0F
000025C0  E310 5010 0014             00000010     1688+         LGF   R1,V2ADDR          load v2 source
000025C6  E761 0000 0806             00000000     1689+         VL    v22,0(R1)          use v22 to test decoder
000025CC  E310 5014 0014             00000014     1690+         LGF   R1,V3ADDR          load v3 source
000025D2  E771 0000 0806             00000000     1691+         VL    v23,0(R1)          use v23 to test decoder
000025D8  E766 7000 3EF2                          1692+         VAVG  V22,V22,V23,3       test instruction (dest is a source)
000025DE  E760 5028 080E             000025A8     1693+         VST   V22,V1O36           save v1 output
000025E4  07FB                                    1694+         BR    R11                return
000025E8                                          1695+RE36     DC    0F                 xl16 expected result
000025E8                                          1696+         DROP  R5
000025E8  80000000 00000002                       1697          DC    XL16'80000000000000028000000000000002'   expected result
000025F0  80000000 00000002
```

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  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000025F8  80000000 00000002                        1698          DC    XL16'80000000000000028000000000000002'   v2
00002600  80000000 00000002
00002608  80000000 00000002                        1699          DC    XL16'80000000000000028000000000000002'   v3
00002610  80000000 00000002
                                                   1700
                                                   1701 * Quadword
                                                   1702          VRR_C VAVG,4
00002618                                           1703+         DS    0FD
00002618                      00002618             1704+         USING *,R5              base for test data and test routine
00002618  00002658                                 1705+T37      DC    A(X37)            address of test routine
0000261C  0025                                     1706+         DC    H'37'             test number
0000261E  00                                       1707+         DC    X'00'
0000261F  04                                       1708+         DC    HL1'4'            m4
00002620  E5C1E5C7 40404040                        1709+         DC    CL8'VAVG'         instruction name
00002628  00002690                                 1710+         DC    A(RE37+16)        address of v2 source
0000262C  000026A0                                 1711+         DC    A(RE37+32)        address of v3 source
00002630  00000010                                 1712+         DC    A(16)             result length
00002634  00002680                                 1713+REA37    DC    A(RE37)           result address
00002638  00000000 00000000                        1714+         DS    FD                gap
00002640  00000000 00000000                        1715+V1O37    DS    XL16              V1 output
00002648  00000000 00000000
00002650  00000000 00000000                        1716+         DS    FD                gap
                                                   1717+*
00002658                                           1718+X37      DS    0F
00002658  E310 5010 0014               00000010    1719+         LGF   R1,V2ADDR         load v2 source
0000265E  E761 0000 0806               00000000    1720+         VL    v22,0(R1)         use v22 to test decoder
00002664  E310 5014 0014               00000014    1721+         LGF   R1,V3ADDR         load v3 source
0000266A  E771 0000 0806               00000000    1722+         VL    v23,0(R1)         use v23 to test decoder
00002670  E766 7000 4EF2                           1723+         VAVG  V22,V22,V23,4      test instruction (dest is a source)
00002676  E760 5028 080E               00002640    1724+         VST   V22,V1O37          save v1 output
0000267C  07FB                                     1725+         BR    R11               return
00002680                                           1726+RE37     DC    0F                xl16 expected result
00002680                                           1727+         DROP  R5
00002680  FF7F7F7F 7F7F7F7F                        1728          DC    XL16'FF7F7F7F 7F7F7F7F 7F7F7F7F 7F7F7F7F'   result
00002688  7F7F7F7F 7F7F7F7F
00002690  7C7C7C7C 7C7C7C7C                        1729          DC    XL16'7C7C7C7C 7C7C7C7C 7C7C7C7C 7C7C7C7C'   v2
00002698  7C7C7C7C 7C7C7C7C
000026A0  82828282 82828282                        1730          DC    XL16'82828282 82828282 82828282 82828282'   v3
000026A8  82828282 82828282
                                                   1731
                                                   1732          VRR_C VAVG,4
000026B0                                           1733+         DS    0FD
000026B0                      000026B0             1734+         USING *,R5              base for test data and test routine
000026B0  000026F0                                 1735+T38      DC    A(X38)            address of test routine
000026B4  0026                                     1736+         DC    H'38'             test number
000026B6  00                                       1737+         DC    X'00'
000026B7  04                                       1738+         DC    HL1'4'            m4
000026B8  E5C1E5C7 40404040                        1739+         DC    CL8'VAVG'         instruction name
000026C0  00002728                                 1740+         DC    A(RE38+16)        address of v2 source
000026C4  00002738                                 1741+         DC    A(RE38+32)        address of v3 source
000026C8  00000010                                 1742+         DC    A(16)             result length
000026CC  00002718                                 1743+REA38    DC    A(RE38)           result address
000026D0  00000000 00000000                        1744+         DS    FD                gap
000026D8  00000000 00000000                        1745+V1O38    DS    XL16              V1 output
000026E0  00000000 00000000
000026E8  00000000 00000000                        1746+         DS    FD                gap
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                  1747+*
000026F0                                          1748+X38      DS    0F
000026F0  E310 5010 0014                00000010  1749+         LGF   R1,V2ADDR          load v2 source
000026F6  E761 0000 0806                00000000  1750+         VL    v22,0(R1)          use v22 to test decoder
000026FC  E310 5014 0014                00000014  1751+         LGF   R1,V3ADDR          load v3 source
00002702  E771 0000 0806                00000000  1752+         VL    v23,0(R1)          use v23 to test decoder
00002708  E766 7000 4EF2                          1753+         VAVG  V22,V22,V23,4       test instruction (dest is a source)
0000270E  E760 5028 080E                000026D8  1754+         VST   V22,V1O38           save v1 output
00002714  07FB                                    1755+         BR    R11                return
00002718                                          1756+RE38     DC    0F                 xl16 expected result
00002718                                          1757+         DROP  R5
00002718  7FFFFFFF FFFFFFFC                       1758          DC    XL16'7FFFFFFF FFFFFFC7 FFFFFFF FFFFFFFC'   result
00002720  7FFFFFFF FFFFFFFC
00002728  7FFFFFFF FFFFFFFC                       1759          DC    XL16'7FFFFFFF FFFFFFC7 FFFFFFF FFFFFFFC'   v2
00002730  7FFFFFFF FFFFFFFC
00002738  7FFFFFFF FFFFFFFC                       1760          DC    XL16'7FFFFFFF FFFFFFC7 FFFFFFF FFFFFFFC'   v3
00002740  7FFFFFFF FFFFFFFC
                                                  1761
                                                  1762          VRR_C VAVG,4
00002748                                          1763+         DS    0FD
00002748                      00002748            1764+         USING *,R5               base for test data and test routine
00002748  00002788                                1765+T39      DC    A(X39)             address of test routine
0000274C  0027                                    1766+         DC    H'39'              test number
0000274E  00                                      1767+         DC    X'00'
0000274F  04                                      1768+         DC    HL1'4'             m4
00002750  E5C1E5C7 40404040                       1769+         DC    CL8'VAVG'          instruction name
00002758  000027C0                                1770+         DC    A(RE39+16)         address of v2 source
0000275C  000027D0                                1771+         DC    A(RE39+32)         address of v3 source
00002760  00000010                                1772+         DC    A(16)              result length
00002764  000027B0                                1773+REA39    DC    A(RE39)            result address
00002768  00000000 00000000                       1774+         DS    FD                 gap
00002770  00000000 00000000                       1775+V1O39    DS    XL16               V1 output
00002778  00000000 00000000
00002780  00000000 00000000                       1776+         DS    FD                 gap
                                                  1777+*
00002788                                          1778+X39      DS    0F
00002788  E310 5010 0014                00000010  1779+         LGF   R1,V2ADDR          load v2 source
0000278E  E761 0000 0806                00000000  1780+         VL    v22,0(R1)          use v22 to test decoder
00002794  E310 5014 0014                00000014  1781+         LGF   R1,V3ADDR          load v3 source
0000279A  E771 0000 0806                00000000  1782+         VL    v23,0(R1)          use v23 to test decoder
000027A0  E766 7000 4EF2                          1783+         VAVG  V22,V22,V23,4       test instruction (dest is a source)
000027A6  E760 5028 080E                00002770  1784+         VST   V22,V1O39           save v1 output
000027AC  07FB                                    1785+         BR    R11                return
000027B0                                          1786+RE39     DC    0F                 xl16 expected result
000027B0                                          1787+         DROP  R5
000027B0  FFFFFFFF FFFFFFFF                       1788          DC    XL16'FFFFFFFF FFFFFFFF 7FFFFFFF FFFFFFFF'   result
000027B8  7FFFFFFF FFFFFFFF
000027C0  80000000 00000002                       1789          DC    XL16'80000000 00000002 80000000 00000002'   v2
000027C8  80000000 00000002
000027D0  7FFFFFFF FFFFFFFC                       1790          DC    XL16'7FFFFFFF FFFFFFFC 7FFFFFFF FFFFFFFC'   v3
000027D8  7FFFFFFF FFFFFFFC
                                                  1791
                                                  1792          VRR_C VAVG,4
000027E0                                          1793+         DS    0FD
000027E0                      000027E0            1794+         USING *,R5               base for test data and test routine
000027E0  00002820                                1795+T40      DC    A(X40)             address of test routine
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000027E4  0028                                     1796+         DC    H'40'               test number
000027E6  00                                       1797+         DC    X'00'
000027E7  04                                       1798+         DC    HL1'4'              m4
000027E8  E5C1E5C7 40404040                        1799+         DC    CL8'VAVG'           instruction name
000027F0  00002858                                 1800+         DC    A(RE40+16)          address of v2 source
000027F4  00002868                                 1801+         DC    A(RE40+32)          address of v3 source
000027F8  00000010                                 1802+         DC    A(16)               result length
000027FC  00002848                                 1803+REA40    DC    A(RE40)             result address
00002800  00000000 00000000                        1804+         DS    FD                  gap
00002808  00000000 00000000                        1805+V1O40    DS    XL16                V1 output
00002810  00000000 00000000
00002818  00000000 00000000                        1806+         DS    FD                  gap
                                                   1807+*
00002820                                           1808+X40      DS    0F
00002820  E310 5010 0014             00000010      1809+         LGF   R1,V2ADDR           load v2 source
00002826  E761 0000 0806             00000000      1810+         VL    v22,0(R1)           use v22 to test decoder
0000282C  E310 5014 0014             00000014      1811+         LGF   R1,V3ADDR           load v3 source
00002832  E771 0000 0806             00000000      1812+         VL    v23,0(R1)           use v23 to test decoder
00002838  E766 7000 4EF2                           1813+         VAVG  V22,V22,V23,4        test instruction (dest is a source)
0000283E  E760 5028 080E             00002808      1814+         VST   V22,V1O40            save v1 output
00002844  07FB                                     1815+         BR    R11                 return
00002848                                           1816+RE40     DC    0F                  xl16 expected result
00002848                                           1817+         DROP  R5
00002848  FFFFFFFF FFFFFFFF                        1818          DC    XL16'FFFFFFFF FFFFFFFF 7FFFFFFF FFFFFFFF'   result
00002850  7FFFFFFF FFFFFFFF
00002858  7FFFFFFF FFFFFFFC                        1819          DC    XL16'7FFFFFFF FFFFFFFC 7FFFFFFF FFFFFFFC'   v2
00002860  7FFFFFFF FFFFFFFC
00002868  80000000 00000002                        1820          DC    XL16'80000000 00000002 80000000 00000002'   v3
00002870  80000000 00000002
                                                   1821
                                                   1822          VRR_C VAVG,4
00002878                                           1823+         DS    0FD
00002878                   00002878                1824+         USING *,R5                base for test data and test routine
00002878  000028B8                                 1825+T41      DC    A(X41)              address of test routine
0000287C  0029                                     1826+         DC    H'41'               test number
0000287E  00                                       1827+         DC    X'00'
0000287F  04                                       1828+         DC    HL1'4'              m4
00002880  E5C1E5C7 40404040                        1829+         DC    CL8'VAVG'           instruction name
00002888  000028F0                                 1830+         DC    A(RE41+16)          address of v2 source
0000288C  00002900                                 1831+         DC    A(RE41+32)          address of v3 source
00002890  00000010                                 1832+         DC    A(16)               result length
00002894  000028E0                                 1833+REA41    DC    A(RE41)             result address
00002898  00000000 00000000                        1834+         DS    FD                  gap
000028A0  00000000 00000000                        1835+V1O41    DS    XL16                V1 output
000028A8  00000000 00000000
000028B0  00000000 00000000                        1836+         DS    FD                  gap
                                                   1837+*
000028B8                                           1838+X41      DS    0F
000028B8  E310 5010 0014             00000010      1839+         LGF   R1,V2ADDR           load v2 source
000028BE  E761 0000 0806             00000000      1840+         VL    v22,0(R1)           use v22 to test decoder
000028C4  E310 5014 0014             00000014      1841+         LGF   R1,V3ADDR           load v3 source
000028CA  E771 0000 0806             00000000      1842+         VL    v23,0(R1)           use v23 to test decoder
000028D0  E766 7000 4EF2                           1843+         VAVG  V22,V22,V23,4        test instruction (dest is a source)
000028D6  E760 5028 080E             000028A0      1844+         VST   V22,V1O41            save v1 output
000028DC  07FB                                     1845+         BR    R11                 return
000028E0                                           1846+RE41     DC    0F                  xl16 expected result
```

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  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

000028E0                                          1847+          DROP  R5
000028E0  80000000 00000002                       1848           DC    XL16'80000000 00000002 80000000 00000002'   result
000028E8  80000000 00000002
000028F0  80000000 00000002                       1849           DC    XL16'80000000 00000002 80000000 00000002'   v2
000028F8  80000000 00000002
00002900  80000000 00000002                       1850           DC    XL16'80000000 00000002 80000000 00000002'   v3
00002908  80000000 00000002
                                                  1851
                                                  1852           VRR_C VAVG,4
00002910                                          1853+          DS    0FD
00002910                     00002910             1854+          USING *,R5                base for test data and test routine
00002910  00002950                                1855+T42       DC    A(X42)              address of test routine
00002914  002A                                    1856+          DC    H'42'               test number
00002916  00                                      1857+          DC    X'00'
00002917  04                                      1858+          DC    HL1'4'              m4
00002918  E5C1E5C7 40404040                       1859+          DC    CL8'VAVG'           instruction name
00002920  00002988                                1860+          DC    A(RE42+16)          address of v2 source
00002924  00002998                                1861+          DC    A(RE42+32)          address of v3 source
00002928  00000010                                1862+          DC    A(16)               result length
0000292C  00002978                                1863+REA42     DC    A(RE42)             result address
00002930  00000000 00000000                       1864+          DS    FD                  gap
00002938  00000000 00000000                       1865+V1O42     DS    XL16                V1 output
00002940  00000000 00000000
00002948  00000000 00000000                       1866+          DS    FD                  gap
                                                  1867+*
00002950                                          1868+X42       DS    0F
00002950  E310 5010 0014               00000010   1869+          LGF   R1,V2ADDR           load v2 source
00002956  E761 0000 0806               00000000   1870+          VL    v22,0(R1)           use v22 to test decoder
0000295C  E310 5014 0014               00000014   1871+          LGF   R1,V3ADDR           load v3 source
00002962  E771 0000 0806               00000000   1872+          VL    v23,0(R1)           use v23 to test decoder
00002968  E766 7000 4EF2                          1873+          VAVG  V22,V22,V23,4        test instruction (dest is a source)
0000296E  E760 5028 080E               00002938   1874+          VST   V22,V1O42            save v1 output
00002974  07FB                                    1875+          BR    R11                 return
00002978                                          1876+RE42      DC    0F                  xl16 expected result
00002978                                          1877+          DROP  R5
00002978  FFFFFFFF FFFFFFFF                       1878           DC    XL16'FFFFFFFF FFFFFFFF 7FFFFFFF FFFFFFFF'   result
00002980  7FFFFFFF FFFFFFFF
00002988  80000000 00000002                       1879           DC    XL16'80000000 00000002 80000000 00000002'   v2
00002990  80000000 00000002
00002998  7FFFFFFF FFFFFFFC                       1880           DC    XL16'7FFFFFFF FFFFFFFC 7FFFFFFF FFFFFFFC'   v3
000029A0  7FFFFFFF FFFFFFFC
                                                  1881
                                                  1882 *--------------------------------------------------------------------
                                                  1883 *VAVGL  - VECTOR AVERAGE LOGICAL
                                                  1884 *--------------------------------------------------------------------
                                                  1885 * Byte
                                                  1886           VRR_C VAVGL,0
000029A8                                          1887+          DS    0FD
000029A8                     000029A8             1888+          USING *,R5                base for test data and test routine
000029A8  000029E8                                1889+T43       DC    A(X43)              address of test routine
000029AC  002B                                    1890+          DC    H'43'               test number
000029AE  00                                      1891+          DC    X'00'
000029AF  00                                      1892+          DC    HL1'0'              m4
000029B0  E5C1E5C7 D3404040                       1893+          DC    CL8'VAVGL'          instruction name
000029B8  00002A20                                1894+          DC    A(RE43+16)          address of v2 source
000029BC  00002A30                                1895+          DC    A(RE43+32)          address of v3 source
```

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  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000029C0  00000010                                 1896+          DC    A(16)              result length
000029C4  00002A10                                 1897+REA43     DC    A(RE43)            result address
000029C8  00000000 00000000                        1898+          DS    FD                 gap
000029D0  00000000 00000000                        1899+V1O43     DS    XL16               V1 output
000029D8  00000000 00000000
000029E0  00000000 00000000                        1900+          DS    FD                 gap
                                                   1901+*
000029E8                                           1902+X43       DS    0F
000029E8  E310 5010 0014                 00000010  1903+          LGF   R1,V2ADDR          load v2 source
000029EE  E761 0000 0806                 00000000  1904+          VL    v22,0(R1)          use v22 to test decoder
000029F4  E310 5014 0014                 00000014  1905+          LGF   R1,V3ADDR          load v3 source
000029FA  E771 0000 0806                 00000000  1906+          VL    v23,0(R1)          use v23 to test decoder
00002A00  E766 7000 0EF0                           1907+          VAVGL V22,V22,V23,0       test instruction (dest is a source)
00002A06  E760 5028 080E                 000029D0  1908+          VST   V22,V1O43           save v1 output
00002A0C  07FB                                     1909+          BR    R11                return
00002A10                                           1910+RE43      DC    0F                 xl16 expected result
00002A10                                           1911+          DROP  R5
00002A10  7F7F7F7F 7F7F7F7F                        1912           DC    XL16'7F7F7F7F7F7F7F7F7F7F7F7F7F7F7F7F'   expected result
00002A18  7F7F7F7F 7F7F7F7F
00002A20  7C7C7C7C 7C7C7C7C                        1913           DC    XL16'7C7C7C7C7C7C7C7C7C7C7C7C7C7C7C7C'   v2
00002A28  7C7C7C7C 7C7C7C7C
00002A30  82828282 82828282                        1914           DC    XL16'82828282828282828282828282828282'   v3
00002A38  82828282 82828282
                                                   1915
                                                   1916 * Halfword
                                                   1917           VRR_C VAVGL,1
00002A40                                           1918+          DS    0FD
00002A40                     00002A40              1919+          USING *,R5               base for test data and test routine
00002A40  00002A80                                 1920+T44       DC    A(X44)             address of test routine
00002A44  002C                                     1921+          DC    H'44'              test number
00002A46  00                                       1922+          DC    X'00'
00002A47  01                                       1923+          DC    HL1'1'             m4
00002A48  E5C1E5C7 D3404040                        1924+          DC    CL8'VAVGL'         instruction name
00002A50  00002AB8                                 1925+          DC    A(RE44+16)         address of v2 source
00002A54  00002AC8                                 1926+          DC    A(RE44+32)         address of v3 source
00002A58  00000010                                 1927+          DC    A(16)              result length
00002A5C  00002AA8                                 1928+REA44     DC    A(RE44)            result address
00002A60  00000000 00000000                        1929+          DS    FD                 gap
00002A68  00000000 00000000                        1930+V1O44     DS    XL16               V1 output
00002A70  00000000 00000000
00002A78  00000000 00000000                        1931+          DS    FD                 gap
                                                   1932+*
00002A80                                           1933+X44       DS    0F
00002A80  E310 5010 0014                 00000010  1934+          LGF   R1,V2ADDR          load v2 source
00002A86  E761 0000 0806                 00000000  1935+          VL    v22,0(R1)          use v22 to test decoder
00002A8C  E310 5014 0014                 00000014  1936+          LGF   R1,V3ADDR          load v3 source
00002A92  E771 0000 0806                 00000000  1937+          VL    v23,0(R1)          use v23 to test decoder
00002A98  E766 7000 1EF0                           1938+          VAVGL V22,V22,V23,1       test instruction (dest is a source)
00002A9E  E760 5028 080E                 00002A68  1939+          VST   V22,V1O44           save v1 output
00002AA4  07FB                                     1940+          BR    R11                return
00002AA8                                           1941+RE44      DC    0F                 xl16 expected result
00002AA8                                           1942+          DROP  R5
00002AA8  7F7F7F7F 7F7F7F7F                        1943           DC    XL16'7F7F7F7F7F7F7F7F7F7F7F7F7F7F7F7F'   expected result
00002AB0  7F7F7F7F 7F7F7F7F
00002AB8  7C7C7C7C 7C7C7C7C                        1944           DC    XL16'7C7C7C7C7C7C7C7C7C7C7C7C7C7C7C7C'   v2
00002AC0  7C7C7C7C 7C7C7C7C
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  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00002AC8  82828282 82828282                       1945          DC    XL16'82828282828282828282828282828282'   v3
00002AD0  82828282 82828282
                                                  1946
                                                  1947 * Word
                                                  1948          VRR_C VAVGL,2
00002AD8                                          1949+         DS    0FD
00002AD8                      00002AD8            1950+         USING *,R5                base for test data and test routine
00002AD8  00002B18                                1951+T45      DC    A(X45)              address of test routine
00002ADC  002D                                    1952+         DC    H'45'               test number
00002ADE  00                                      1953+         DC    X'00'
00002ADF  02                                      1954+         DC    HL1'2'              m4
00002AE0  E5C1E5C7 D3404040                       1955+         DC    CL8'VAVGL'          instruction name
00002AE8  00002B50                                1956+         DC    A(RE45+16)          address of v2 source
00002AEC  00002B60                                1957+         DC    A(RE45+32)          address of v3 source
00002AF0  00000010                                1958+         DC    A(16)               result length
00002AF4  00002B40                                1959+REA45    DC    A(RE45)             result address
00002AF8  00000000 00000000                       1960+         DS    FD                  gap
00002B00  00000000 00000000                       1961+V1O45    DS    XL16                V1 output
00002B08  00000000 00000000
00002B10  00000000 00000000                       1962+         DS    FD                  gap
                                                  1963+*
00002B18                                          1964+X45      DS    0F
00002B18  E310 5010 0014             00000010     1965+         LGF   R1,V2ADDR           load v2 source
00002B1E  E761 0000 0806             00000000     1966+         VL    v22,0(R1)           use v22 to test decoder
00002B24  E310 5014 0014             00000014     1967+         LGF   R1,V3ADDR           load v3 source
00002B2A  E771 0000 0806             00000000     1968+         VL    v23,0(R1)           use v23 to test decoder
00002B30  E766 7000 2EF0                          1969+         VAVGL V22,V22,V23,2        test instruction (dest is a source)
00002B36  E760 5028 080E             00002B00     1970+         VST   V22,V1O45            save v1 output
00002B3C  07FB                                    1971+         BR    R11                 return
00002B40                                          1972+RE45     DC    0F                  xl16 expected result
00002B40                                          1973+         DROP  R5
00002B40  7F7F7F7F 7F7F7F7F                       1974          DC    XL16'7F7F7F7F7F7F7F7F7F7F7F7F7F7F7F7F'   expected result
00002B48  7F7F7F7F 7F7F7F7F
00002B50  7C7C7C7C 7C7C7C7C                       1975          DC    XL16'7C7C7C7C7C7C7C7C7C7C7C7C7C7C7C7C'   v2
00002B58  7C7C7C7C 7C7C7C7C
00002B60  82828282 82828282                       1976          DC    XL16'82828282828282828282828282828282'   v3
00002B68  82828282 82828282
                                                  1977
                                                  1978 * Doubleword
                                                  1979          VRR_C VAVGL,3
00002B70                                          1980+         DS    0FD
00002B70                      00002B70            1981+         USING *,R5                base for test data and test routine
00002B70  00002BB0                                1982+T46      DC    A(X46)              address of test routine
00002B74  002E                                    1983+         DC    H'46'               test number
00002B76  00                                      1984+         DC    X'00'
00002B77  03                                      1985+         DC    HL1'3'              m4
00002B78  E5C1E5C7 D3404040                       1986+         DC    CL8'VAVGL'          instruction name
00002B80  00002BE8                                1987+         DC    A(RE46+16)          address of v2 source
00002B84  00002BF8                                1988+         DC    A(RE46+32)          address of v3 source
00002B88  00000010                                1989+         DC    A(16)               result length
00002B8C  00002BD8                                1990+REA46    DC    A(RE46)             result address
00002B90  00000000 00000000                       1991+         DS    FD                  gap
00002B98  00000000 00000000                       1992+V1O46    DS    XL16                V1 output
00002BA0  00000000 00000000
00002BA8  00000000 00000000                       1993+         DS    FD                  gap
                                                  1994+*
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  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002BB0                                           1995+X46       DS    0F
00002BB0  E310 5010 0014                00000010   1996+          LGF   R1,V2ADDR         load v2 source
00002BB6  E761 0000 0806                00000000   1997+          VL    v22,0(R1)         use v22 to test decoder
00002BBC  E310 5014 0014                00000014   1998+          LGF   R1,V3ADDR         load v3 source
00002BC2  E771 0000 0806                00000000   1999+          VL    v23,0(R1)         use v23 to test decoder
00002BC8  E766 7000 3EF0                           2000+          VAVGL V22,V22,V23,3      test instruction (dest is a source)
00002BCE  E760 5028 080E                00002B98   2001+          VST   V22,V1O46          save v1 output
00002BD4  07FB                                     2002+          BR    R11               return
00002BD8                                           2003+RE46      DC    0F                xl16 expected result
00002BD8                                           2004+          DROP  R5
00002BD8  7F7F7F7F 7F7F7F7F                        2005           DC    XL16'7F7F7F7F7F7F7F7F7F7F7F7F7F7F7F7F'   expected result
00002BE0  7F7F7F7F 7F7F7F7F
00002BE8  7C7C7C7C 7C7C7C7C                        2006           DC    XL16'7C7C7C7C7C7C7C7C7C7C7C7C7C7C7C7C'   v2
00002BF0  7C7C7C7C 7C7C7C7C
00002BF8  82828282 82828282                        2007           DC    XL16'82828282828282828282828282828282'   v3
00002C00  82828282 82828282
                                                   2008
                                                   2009 * Doubleword
                                                   2010           VRR_C VAVGL,3
00002C08                                           2011+          DS    0FD
00002C08                      00002C08             2012+          USING *,R5              base for test data and test routine
00002C08  00002C48                                 2013+T47       DC    A(X47)            address of test routine
00002C0C  002F                                     2014+          DC    H'47'             test number
00002C0E  00                                       2015+          DC    X'00'
00002C0F  03                                       2016+          DC    HL1'3'            m4
00002C10  E5C1E5C7 D3404040                        2017+          DC    CL8'VAVGL'        instruction name
00002C18  00002C80                                 2018+          DC    A(RE47+16)        address of v2 source
00002C1C  00002C90                                 2019+          DC    A(RE47+32)        address of v3 source
00002C20  00000010                                 2020+          DC    A(16)             result length
00002C24  00002C70                                 2021+REA47     DC    A(RE47)           result address
00002C28  00000000 00000000                        2022+          DS    FD                gap
00002C30  00000000 00000000                        2023+V1O47     DS    XL16              V1 output
00002C38  00000000 00000000
00002C40  00000000 00000000                        2024+          DS    FD                gap
                                                   2025+*
00002C48                                           2026+X47       DS    0F
00002C48  E310 5010 0014                00000010   2027+          LGF   R1,V2ADDR         load v2 source
00002C4E  E761 0000 0806                00000000   2028+          VL    v22,0(R1)         use v22 to test decoder
00002C54  E310 5014 0014                00000014   2029+          LGF   R1,V3ADDR         load v3 source
00002C5A  E771 0000 0806                00000000   2030+          VL    v23,0(R1)         use v23 to test decoder
00002C60  E766 7000 3EF0                           2031+          VAVGL V22,V22,V23,3      test instruction (dest is a source)
00002C66  E760 5028 080E                00002C30   2032+          VST   V22,V1O47          save v1 output
00002C6C  07FB                                     2033+          BR    R11               return
00002C70                                           2034+RE47      DC    0F                xl16 expected result
00002C70                                           2035+          DROP  R5
00002C70  7FFFFFFF FFFFFFFC                        2036           DC    XL16'7FFFFFFFFFFFFFFC7FFFFFFFFFFFFFFC'   expected result
00002C78  7FFFFFFF FFFFFFFC
00002C80  7FFFFFFF FFFFFFFC                        2037           DC    XL16'7FFFFFFFFFFFFFFC7FFFFFFFFFFFFFFC'   v2
00002C88  7FFFFFFF FFFFFFFC
00002C90  7FFFFFFF FFFFFFFC                        2038           DC    XL16'7FFFFFFFFFFFFFFC7FFFFFFFFFFFFFFC'   v3
00002C98  7FFFFFFF FFFFFFFC
                                                   2039
                                                   2040 * Doubleword
                                                   2041           VRR_C VAVGL,3
00002CA0                                           2042+          DS    0FD
00002CA0                      00002CA0             2043+          USING *,R5              base for test data and test routine
```

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  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002CA0  00002CE0                               2044+T48      DC    A(X48)              address of test routine
00002CA4  0030                                   2045+         DC    H'48'               test number
00002CA6  00                                     2046+         DC    X'00'
00002CA7  03                                     2047+         DC    HL1'3'              m4
00002CA8  E5C1E5C7 D3404040                      2048+         DC    CL8'VAVGL'          instruction name
00002CB0  00002D18                               2049+         DC    A(RE48+16)          address of v2 source
00002CB4  00002D28                               2050+         DC    A(RE48+32)          address of v3 source
00002CB8  00000010                               2051+         DC    A(16)               result length
00002CBC  00002D08                               2052+REA48    DC    A(RE48)             result address
00002CC0  00000000 00000000                      2053+         DS    FD                  gap
00002CC8  00000000 00000000                      2054+V1O48    DS    XL16                V1 output
00002CD0  00000000 00000000
00002CD8  00000000 00000000                      2055+         DS    FD                  gap
                                                 2056+*
00002CE0                                         2057+X48      DS    0F
00002CE0  E310 5010 0014             00000010    2058+         LGF   R1,V2ADDR           load v2 source
00002CE6  E761 0000 0806             00000000    2059+         VL    v22,0(R1)           use v22 to test decoder
00002CEC  E310 5014 0014             00000014    2060+         LGF   R1,V3ADDR           load v3 source
00002CF2  E771 0000 0806             00000000    2061+         VL    v23,0(R1)           use v23 to test decoder
00002CF8  E766 7000 3EF0                         2062+         VAVGL V22,V22,V23,3        test instruction (dest is a source)
00002CFE  E760 5028 080E             00002CC8    2063+         VST   V22,V1O48            save v1 output
00002D04  07FB                                   2064+         BR    R11                 return
00002D08                                         2065+RE48     DC    0F                  xl16 expected result
00002D08                                         2066+         DROP  R5
00002D08  7FFFFFFF FFFFFFFF                      2067          DC    XL16'7FFFFFFFFFFFFFFF7FFFFFFFFFFFFFFF'   expected result
00002D10  7FFFFFFF FFFFFFFF
00002D18  80000000 00000002                      2068          DC    XL16'80000000000000028000000000000002'   v2
00002D20  80000000 00000002
00002D28  7FFFFFFF FFFFFFFC                      2069          DC    XL16'7FFFFFFFFFFFFFFC7FFFFFFFFFFFFFFC'   v3
00002D30  7FFFFFFF FFFFFFFC
                                                 2070
                                                 2071 * Doubleword
                                                 2072          VRR_C VAVGL,3
00002D38                                         2073+         DS    0FD
00002D38                       00002D38          2074+         USING *,R5              base for test data and test routine
00002D38  00002D78                               2075+T49      DC    A(X49)              address of test routine
00002D3C  0031                                   2076+         DC    H'49'               test number
00002D3E  00                                     2077+         DC    X'00'
00002D3F  03                                     2078+         DC    HL1'3'              m4
00002D40  E5C1E5C7 D3404040                      2079+         DC    CL8'VAVGL'          instruction name
00002D48  00002DB0                               2080+         DC    A(RE49+16)          address of v2 source
00002D4C  00002DC0                               2081+         DC    A(RE49+32)          address of v3 source
00002D50  00000010                               2082+         DC    A(16)               result length
00002D54  00002DA0                               2083+REA49    DC    A(RE49)             result address
00002D58  00000000 00000000                      2084+         DS    FD                  gap
00002D60  00000000 00000000                      2085+V1O49    DS    XL16                V1 output
00002D68  00000000 00000000
00002D70  00000000 00000000                      2086+         DS    FD                  gap
                                                 2087+*
00002D78                                         2088+X49      DS    0F
00002D78  E310 5010 0014             00000010    2089+         LGF   R1,V2ADDR           load v2 source
00002D7E  E761 0000 0806             00000000    2090+         VL    v22,0(R1)           use v22 to test decoder
00002D84  E310 5014 0014             00000014    2091+         LGF   R1,V3ADDR           load v3 source
00002D8A  E771 0000 0806             00000000    2092+         VL    v23,0(R1)           use v23 to test decoder
00002D90  E766 7000 3EF0                         2093+         VAVGL V22,V22,V23,3        test instruction (dest is a source)
00002D96  E760 5028 080E             00002D60    2094+         VST   V22,V1O49            save v1 output
```

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  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002D9C  07FB                                     2095+          BR     R11                 return
00002DA0                                           2096+RE49      DC     0F                  xl16 expected result
00002DA0                                           2097+          DROP   R5
00002DA0  7FFFFFFF FFFFFFFF                        2098           DC     XL16'7FFFFFFFFFFFFFFF7FFFFFFFFFFFFFFF'   expected result
00002DA8  7FFFFFFF FFFFFFFF
00002DB0  7FFFFFFF FFFFFFFC                        2099           DC     XL16'7FFFFFFFFFFFFFFC7FFFFFFFFFFFFFFC'    v2
00002DB8  7FFFFFFF FFFFFFFC
00002DC0  80000000 00000002                        2100           DC     XL16'80000000000000028000000000000002'   v3
00002DC8  80000000 00000002
                                                   2101
                                                   2102 * Doubleword
                                                   2103           VRR_C VAVGL,3
00002DD0                                           2104+          DS     0FD
00002DD0                       00002DD0            2105+          USING *,R5                 base for test data and test routine
00002DD0  00002E10                                 2106+T50       DC     A(X50)              address of test routine
00002DD4  0032                                     2107+          DC     H'50'               test number
00002DD6  00                                       2108+          DC     X'00'
00002DD7  03                                       2109+          DC     HL1'3'              m4
00002DD8  E5C1E5C7 D3404040                        2110+          DC     CL8'VAVGL'          instruction name
00002DE0  00002E48                                 2111+          DC     A(RE50+16)          address of v2 source
00002DE4  00002E58                                 2112+          DC     A(RE50+32)          address of v3 source
00002DE8  00000010                                 2113+          DC     A(16)               result length
00002DEC  00002E38                                 2114+REA50     DC     A(RE50)             result address
00002DF0  00000000 00000000                        2115+          DS     FD                  gap
00002DF8  00000000 00000000                        2116+V1O50     DS     XL16                V1 output
00002E00  00000000 00000000
00002E08  00000000 00000000                        2117+          DS     FD                  gap
                                                   2118+*
00002E10                                           2119+X50       DS     0F
00002E10  E310 5010 0014                 00000010  2120+          LGF    R1,V2ADDR           load v2 source
00002E16  E761 0000 0806                 00000000  2121+          VL     v22,0(R1)           use v22 to test decoder
00002E1C  E310 5014 0014                 00000014  2122+          LGF    R1,V3ADDR           load v3 source
00002E22  E771 0000 0806                 00000000  2123+          VL     v23,0(R1)           use v23 to test decoder
00002E28  E766 7000 3EF0                           2124+          VAVGL  V22,V22,V23,3        test instruction (dest is a source)
00002E2E  E760 5028 080E                 00002DF8  2125+          VST    V22,V1O50            save v1 output
00002E34  07FB                                     2126+          BR     R11                 return
00002E38                                           2127+RE50      DC     0F                  xl16 expected result
00002E38                                           2128+          DROP   R5
00002E38  80000000 00000002                        2129           DC     XL16'80000000000000028000000000000002'   expected result
00002E40  80000000 00000002
00002E48  80000000 00000002                        2130           DC     XL16'80000000000000028000000000000002'   v2
00002E50  80000000 00000002
00002E58  80000000 00000002                        2131           DC     XL16'80000000000000028000000000000002'   v3
00002E60  80000000 00000002
                                                   2132
                                                   2133
                                                   2134
                                                   2135 * Quadword
                                                   2136           VRR_C VAVGL,4
00002E68                                           2137+          DS     0FD
00002E68                       00002E68            2138+          USING *,R5                 base for test data and test routine
00002E68  00002EA8                                 2139+T51       DC     A(X51)              address of test routine
00002E6C  0033                                     2140+          DC     H'51'               test number
00002E6E  00                                       2141+          DC     X'00'
00002E6F  04                                       2142+          DC     HL1'4'              m4
00002E70  E5C1E5C7 D3404040                        2143+          DC     CL8'VAVGL'          instruction name
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  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002E78  00002EE0                                 2144+          DC    A(RE51+16)        address of v2 source
00002E7C  00002EF0                                 2145+          DC    A(RE51+32)        address of v3 source
00002E80  00000010                                 2146+          DC    A(16)             result length
00002E84  00002ED0                                 2147+REA51     DC    A(RE51)           result address
00002E88  00000000 00000000                        2148+          DS    FD                gap
00002E90  00000000 00000000                        2149+V1O51     DS    XL16              V1 output
00002E98  00000000 00000000
00002EA0  00000000 00000000                        2150+          DS    FD                gap
                                                   2151+*
00002EA8                                           2152+X51       DS    0F
00002EA8  E310 5010 0014                00000010   2153+          LGF   R1,V2ADDR         load v2 source
00002EAE  E761 0000 0806                00000000   2154+          VL    v22,0(R1)         use v22 to test decoder
00002EB4  E310 5014 0014                00000014   2155+          LGF   R1,V3ADDR         load v3 source
00002EBA  E771 0000 0806                00000000   2156+          VL    v23,0(R1)         use v23 to test decoder
00002EC0  E766 7000 4EF0                           2157+          VAVGL V22,V22,V23,4      test instruction (dest is a source)
00002EC6  E760 5028 080E                00002E90   2158+          VST   V22,V1O51          save v1 output
00002ECC  07FB                                     2159+          BR    R11               return
00002ED0                                           2160+RE51      DC    0F                xl16 expected result
00002ED0                                           2161+          DROP  R5
00002ED0  7F7F7F7F 7F7F7F7F                        2162           DC    XL16'7F7F7F7F7F7F7F7F7F7F7F7F7F7F7F7F'   expected result
00002ED8  7F7F7F7F 7F7F7F7F
00002EE0  7C7C7C7C 7C7C7C7C                        2163           DC    XL16'7C7C7C7C7C7C7C7C7C7C7C7C7C7C7C7C'   v2
00002EE8  7C7C7C7C 7C7C7C7C
00002EF0  82828282 82828282                        2164           DC    XL16'82828282828282828282828282828282'   v3
00002EF8  82828282 82828282
                                                   2165
                                                   2166 * Quadword
                                                   2167           VRR_C VAVGL,4
00002F00                                           2168+          DS    0FD
00002F00                      00002F00             2169+          USING *,R5              base for test data and test routine
00002F00  00002F40                                 2170+T52       DC    A(X52)            address of test routine
00002F04  0034                                     2171+          DC    H'52'             test number
00002F06  00                                       2172+          DC    X'00'
00002F07  04                                       2173+          DC    HL1'4'            m4
00002F08  E5C1E5C7 D3404040                        2174+          DC    CL8'VAVGL'        instruction name
00002F10  00002F78                                 2175+          DC    A(RE52+16)        address of v2 source
00002F14  00002F88                                 2176+          DC    A(RE52+32)        address of v3 source
00002F18  00000010                                 2177+          DC    A(16)             result length
00002F1C  00002F68                                 2178+REA52     DC    A(RE52)           result address
00002F20  00000000 00000000                        2179+          DS    FD                gap
00002F28  00000000 00000000                        2180+V1O52     DS    XL16              V1 output
00002F30  00000000 00000000
00002F38  00000000 00000000                        2181+          DS    FD                gap
                                                   2182+*
00002F40                                           2183+X52       DS    0F
00002F40  E310 5010 0014                00000010   2184+          LGF   R1,V2ADDR         load v2 source
00002F46  E761 0000 0806                00000000   2185+          VL    v22,0(R1)         use v22 to test decoder
00002F4C  E310 5014 0014                00000014   2186+          LGF   R1,V3ADDR         load v3 source
00002F52  E771 0000 0806                00000000   2187+          VL    v23,0(R1)         use v23 to test decoder
00002F58  E766 7000 4EF0                           2188+          VAVGL V22,V22,V23,4      test instruction (dest is a source)
00002F5E  E760 5028 080E                00002F28   2189+          VST   V22,V1O52          save v1 output
00002F64  07FB                                     2190+          BR    R11               return
00002F68                                           2191+RE52      DC    0F                xl16 expected result
00002F68                                           2192+          DROP  R5
00002F68  7FFFFFFF FFFFFFFC                        2193           DC    XL16'7FFFFFFFFFFFFFFC7FFFFFFFFFFFFFFC'   expected result
00002F70  7FFFFFFF FFFFFFFC
```

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  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002F78  7FFFFFFF FFFFFFFC                        2194          DC    XL16'7FFFFFFFFFFFFFFC7FFFFFFFFFFFFFFC'   v2
00002F80  7FFFFFFF FFFFFFFC
00002F88  7FFFFFFF FFFFFFFC                        2195          DC    XL16'7FFFFFFFFFFFFFFC7FFFFFFFFFFFFFFC'   v3
00002F90  7FFFFFFF FFFFFFFC
                                                   2196
                                                   2197 * Quadword
                                                   2198          VRR_C VAVGL,4
00002F98                                           2199+         DS    0FD
00002F98                     00002F98              2200+         USING *,R5              base for test data and test routine
00002F98  00002FD8                                 2201+T53      DC    A(X53)            address of test routine
00002F9C  0035                                     2202+         DC    H'53'             test number
00002F9E  00                                       2203+         DC    X'00'
00002F9F  04                                       2204+         DC    HL1'4'            m4
00002FA0  E5C1E5C7 D3404040                        2205+         DC    CL8'VAVGL'        instruction name
00002FA8  00003010                                 2206+         DC    A(RE53+16)        address of v2 source
00002FAC  00003020                                 2207+         DC    A(RE53+32)        address of v3 source
00002FB0  00000010                                 2208+         DC    A(16)             result length
00002FB4  00003000                                 2209+REA53    DC    A(RE53)           result address
00002FB8  00000000 00000000                        2210+         DS    FD                gap
00002FC0  00000000 00000000                        2211+V1O53    DS    XL16              V1 output
00002FC8  00000000 00000000
00002FD0  00000000 00000000                        2212+         DS    FD                gap
                                                   2213+*
00002FD8                                           2214+X53      DS    0F
00002FD8  E310 5010 0014               00000010    2215+         LGF   R1,V2ADDR         load v2 source
00002FDE  E761 0000 0806               00000000    2216+         VL    v22,0(R1)         use v22 to test decoder
00002FE4  E310 5014 0014               00000014    2217+         LGF   R1,V3ADDR         load v3 source
00002FEA  E771 0000 0806               00000000    2218+         VL    v23,0(R1)         use v23 to test decoder
00002FF0  E766 7000 4EF0                           2219+         VAVGL V22,V22,V23,4      test instruction (dest is a source)
00002FF6  E760 5028 080E               00002FC0    2220+         VST   V22,V1O53          save v1 output
00002FFC  07FB                                     2221+         BR    R11               return
00003000                                           2222+RE53     DC    0F                xl16 expected result
00003000                                           2223+         DROP  R5
00003000  7FFFFFFF FFFFFFFF                        2224          DC    XL16'7FFFFFFFFFFFFFFF7FFFFFFFFFFFFFFF'   expected result
00003008  7FFFFFFF FFFFFFFF
00003010  80000000 00000002                        2225          DC    XL16'80000000000000028000000000000002'   v2
00003018  80000000 00000002
00003020  7FFFFFFF FFFFFFFC                        2226          DC    XL16'7FFFFFFFFFFFFFFC7FFFFFFFFFFFFFFC'   v3
00003028  7FFFFFFF FFFFFFFC
                                                   2227
                                                   2228 * Quadword
                                                   2229          VRR_C VAVGL,4
00003030                                           2230+         DS    0FD
00003030                     00003030              2231+         USING *,R5              base for test data and test routine
00003030  00003070                                 2232+T54      DC    A(X54)            address of test routine
00003034  0036                                     2233+         DC    H'54'             test number
00003036  00                                       2234+         DC    X'00'
00003037  04                                       2235+         DC    HL1'4'            m4
00003038  E5C1E5C7 D3404040                        2236+         DC    CL8'VAVGL'        instruction name
00003040  000030A8                                 2237+         DC    A(RE54+16)        address of v2 source
00003044  000030B8                                 2238+         DC    A(RE54+32)        address of v3 source
00003048  00000010                                 2239+         DC    A(16)             result length
0000304C  00003098                                 2240+REA54    DC    A(RE54)           result address
00003050  00000000 00000000                        2241+         DS    FD                gap
00003058  00000000 00000000                        2242+V1O54    DS    XL16              V1 output
00003060  00000000 00000000
```

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  LOC       OBJECT CODE       ADDR1     ADDR2     STMT

00003068  00000000 00000000                      2243+         DS    FD                 gap
                                                 2244+*
00003070                                         2245+X54      DS    0F
00003070  E310 5010 0014               00000010  2246+         LGF   R1,V2ADDR          load v2 source
00003076  E761 0000 0806               00000000  2247+         VL    v22,0(R1)          use v22 to test decoder
0000307C  E310 5014 0014               00000014  2248+         LGF   R1,V3ADDR          load v3 source
00003082  E771 0000 0806               00000000  2249+         VL    v23,0(R1)          use v23 to test decoder
00003088  E766 7000 4EF0                         2250+         VAVGL V22,V22,V23,4       test instruction (dest is a source)
0000308E  E760 5028 080E               00003058  2251+         VST   V22,V1O54           save v1 output
00003094  07FB                                   2252+         BR    R11                return
00003098                                         2253+RE54     DC    0F                 xl16 expected result
00003098                                         2254+         DROP  R5
00003098  7FFFFFFF FFFFFFFF                      2255          DC    XL16'7FFFFFFFFFFFFFFF7FFFFFFFFFFFFFFF'   expected result
000030A0  7FFFFFFF FFFFFFFF
000030A8  7FFFFFFF FFFFFFFC                      2256          DC    XL16'7FFFFFFFFFFFFFFC7FFFFFFFFFFFFFFC'   v2
000030B0  7FFFFFFF FFFFFFFC
000030B8  80000000 00000002                      2257          DC    XL16'80000000000000028000000000000002'   v3
000030C0  80000000 00000002
                                                 2258
                                                 2259 * Quadword
                                                 2260          VRR_C VAVGL,4
000030C8                                         2261+         DS    0FD
000030C8                    000030C8             2262+         USING *,R5               base for test data and test routine
000030C8  00003108                               2263+T55      DC    A(X55)             address of test routine
000030CC  0037                                   2264+         DC    H'55'              test number
000030CE  00                                     2265+         DC    X'00'
000030CF  04                                     2266+         DC    HL1'4'             m4
000030D0  E5C1E5C7 D3404040                      2267+         DC    CL8'VAVGL'         instruction name
000030D8  00003140                               2268+         DC    A(RE55+16)         address of v2 source
000030DC  00003150                               2269+         DC    A(RE55+32)         address of v3 source
000030E0  00000010                               2270+         DC    A(16)              result length
000030E4  00003130                               2271+REA55    DC    A(RE55)            result address
000030E8  00000000 00000000                      2272+         DS    FD                 gap
000030F0  00000000 00000000                      2273+V1O55    DS    XL16               V1 output
000030F8  00000000 00000000
00003100  00000000 00000000                      2274+         DS    FD                 gap
                                                 2275+*
00003108                                         2276+X55      DS    0F
00003108  E310 5010 0014               00000010  2277+         LGF   R1,V2ADDR          load v2 source
0000310E  E761 0000 0806               00000000  2278+         VL    v22,0(R1)          use v22 to test decoder
00003114  E310 5014 0014               00000014  2279+         LGF   R1,V3ADDR          load v3 source
0000311A  E771 0000 0806               00000000  2280+         VL    v23,0(R1)          use v23 to test decoder
00003120  E766 7000 4EF0                         2281+         VAVGL V22,V22,V23,4       test instruction (dest is a source)
00003126  E760 5028 080E               000030F0  2282+         VST   V22,V1O55           save v1 output
0000312C  07FB                                   2283+         BR    R11                return
00003130                                         2284+RE55     DC    0F                 xl16 expected result
00003130                                         2285+         DROP  R5
00003130  80000000 00000002                      2286          DC    XL16'80000000000000028000000000000002'   expected result
00003138  80000000 00000002
00003140  80000000 00000002                      2287          DC    XL16'80000000000000028000000000000002'   v2
00003148  80000000 00000002
00003150  80000000 00000002                      2288          DC    XL16'80000000000000028000000000000002'   v3
00003158  80000000 00000002
                                                 2289
                                                 2290
                                                 2291
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00003160  00000000                                 2292          DC    F'0'     END OF TABLE
00003164  00000000                                 2293          DC    F'0'
                                                   2294 *
                                                   2295 * table of pointers to individual load test
                                                   2296 *
00003168                                           2297 E7TESTS  DS    0F
                                                   2298          PTTABLE
00003168                                           2299+TTABLE   DS    0F
00003168  000010B8                                 2300+         DC    A(T1)             TEST &CUR
0000316C  00001150                                 2301+         DC    A(T2)             TEST &CUR
00003170  000011E8                                 2302+         DC    A(T3)             TEST &CUR
00003174  00001280                                 2303+         DC    A(T4)             TEST &CUR
00003178  00001318                                 2304+         DC    A(T5)             TEST &CUR
0000317C  000013B0                                 2305+         DC    A(T6)             TEST &CUR
00003180  00001448                                 2306+         DC    A(T7)             TEST &CUR
00003184  000014E0                                 2307+         DC    A(T8)             TEST &CUR
00003188  00001578                                 2308+         DC    A(T9)             TEST &CUR
0000318C  00001610                                 2309+         DC    A(T10)            TEST &CUR
00003190  000016A8                                 2310+         DC    A(T11)            TEST &CUR
00003194  00001740                                 2311+         DC    A(T12)            TEST &CUR
00003198  000017D8                                 2312+         DC    A(T13)            TEST &CUR
0000319C  00001870                                 2313+         DC    A(T14)            TEST &CUR
000031A0  00001908                                 2314+         DC    A(T15)            TEST &CUR
000031A4  000019A0                                 2315+         DC    A(T16)            TEST &CUR
000031A8  00001A38                                 2316+         DC    A(T17)            TEST &CUR
000031AC  00001AD0                                 2317+         DC    A(T18)            TEST &CUR
000031B0  00001B68                                 2318+         DC    A(T19)            TEST &CUR
000031B4  00001C00                                 2319+         DC    A(T20)            TEST &CUR
000031B8  00001C98                                 2320+         DC    A(T21)            TEST &CUR
000031BC  00001D30                                 2321+         DC    A(T22)            TEST &CUR
000031C0  00001DC8                                 2322+         DC    A(T23)            TEST &CUR
000031C4  00001E60                                 2323+         DC    A(T24)            TEST &CUR
000031C8  00001EF8                                 2324+         DC    A(T25)            TEST &CUR
000031CC  00001F90                                 2325+         DC    A(T26)            TEST &CUR
000031D0  00002028                                 2326+         DC    A(T27)            TEST &CUR
000031D4  000020C0                                 2327+         DC    A(T28)            TEST &CUR
000031D8  00002158                                 2328+         DC    A(T29)            TEST &CUR
000031DC  000021F0                                 2329+         DC    A(T30)            TEST &CUR
000031E0  00002288                                 2330+         DC    A(T31)            TEST &CUR
000031E4  00002320                                 2331+         DC    A(T32)            TEST &CUR
000031E8  000023B8                                 2332+         DC    A(T33)            TEST &CUR
000031EC  00002450                                 2333+         DC    A(T34)            TEST &CUR
000031F0  000024E8                                 2334+         DC    A(T35)            TEST &CUR
000031F4  00002580                                 2335+         DC    A(T36)            TEST &CUR
000031F8  00002618                                 2336+         DC    A(T37)            TEST &CUR
000031FC  000026B0                                 2337+         DC    A(T38)            TEST &CUR
00003200  00002748                                 2338+         DC    A(T39)            TEST &CUR
00003204  000027E0                                 2339+         DC    A(T40)            TEST &CUR
00003208  00002878                                 2340+         DC    A(T41)            TEST &CUR
0000320C  00002910                                 2341+         DC    A(T42)            TEST &CUR
00003210  000029A8                                 2342+         DC    A(T43)            TEST &CUR
00003214  00002A40                                 2343+         DC    A(T44)            TEST &CUR
00003218  00002AD8                                 2344+         DC    A(T45)            TEST &CUR
0000321C  00002B70                                 2345+         DC    A(T46)            TEST &CUR
00003220  00002C08                                 2346+         DC    A(T47)            TEST &CUR
00003224  00002CA0                                 2347+         DC    A(T48)            TEST &CUR
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00003228  00002D38                                2348+         DC    A(T49)              TEST &CUR
0000322C  00002DD0                                2349+         DC    A(T50)              TEST &CUR
00003230  00002E68                                2350+         DC    A(T51)              TEST &CUR
00003234  00002F00                                2351+         DC    A(T52)              TEST &CUR
00003238  00002F98                                2352+         DC    A(T53)              TEST &CUR
0000323C  00003030                                2353+         DC    A(T54)              TEST &CUR
00003240  000030C8                                2354+         DC    A(T55)              TEST &CUR
                                                  2355+*
00003244  00000000                                2356+         DC    A(0)                END OF TABLE
00003248  00000000                                2357+         DC    A(0)
                                                  2358
0000324C  00000000                                2359          DC    F'0'     END OF TABLE
00003250  00000000                                2360          DC    F'0'
```

```
  LOC        OBJECT CODE        ADDR1     ADDR2    STMT

                                                   2362 ********************************************************************
                                                   2363 *        Register equates
                                                   2364 ********************************************************************

                                00000000  00000001  2366 R0       EQU   0
                                00000001  00000001  2367 R1       EQU   1
                                00000002  00000001  2368 R2       EQU   2
                                00000003  00000001  2369 R3       EQU   3
                                00000004  00000001  2370 R4       EQU   4
                                00000005  00000001  2371 R5       EQU   5
                                00000006  00000001  2372 R6       EQU   6
                                00000007  00000001  2373 R7       EQU   7
                                00000008  00000001  2374 R8       EQU   8
                                00000009  00000001  2375 R9       EQU   9
                                0000000A  00000001  2376 R10      EQU   10
                                0000000B  00000001  2377 R11      EQU   11
                                0000000C  00000001  2378 R12      EQU   12
                                0000000D  00000001  2379 R13      EQU   13
                                0000000E  00000001  2380 R14      EQU   14
                                0000000F  00000001  2381 R15      EQU   15




                                                   2383 ********************************************************************
                                                   2384 *        Register equates
                                                   2385 ********************************************************************

                                00000000  00000001  2387 V0       EQU   0
                                00000001  00000001  2388 V1       EQU   1
                                00000002  00000001  2389 V2       EQU   2
                                00000003  00000001  2390 V3       EQU   3
                                00000004  00000001  2391 V4       EQU   4
                                00000005  00000001  2392 V5       EQU   5
                                00000006  00000001  2393 V6       EQU   6
                                00000007  00000001  2394 V7       EQU   7
                                00000008  00000001  2395 V8       EQU   8
                                00000009  00000001  2396 V9       EQU   9
                                0000000A  00000001  2397 V10      EQU   10
                                0000000B  00000001  2398 V11      EQU   11
                                0000000C  00000001  2399 V12      EQU   12
                                0000000D  00000001  2400 V13      EQU   13
                                0000000E  00000001  2401 V14      EQU   14
                                0000000F  00000001  2402 V15      EQU   15
                                00000010  00000001  2403 V16      EQU   16
                                00000011  00000001  2404 V17      EQU   17
                                00000012  00000001  2405 V18      EQU   18
                                00000013  00000001  2406 V19      EQU   19
                                00000014  00000001  2407 V20      EQU   20
                                00000015  00000001  2408 V21      EQU   21
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                             00000016  00000001  2409 V22       EQU   22
                             00000017  00000001  2410 V23       EQU   23
                             00000018  00000001  2411 V24       EQU   24
                             00000019  00000001  2412 V25       EQU   25
                             0000001A  00000001  2413 V26       EQU   26
                             0000001B  00000001  2414 V27       EQU   27
                             0000001C  00000001  2415 V28       EQU   28
                             0000001D  00000001  2416 V29       EQU   29
                             0000001E  00000001  2417 V30       EQU   30
                             0000001F  00000001  2418 V31       EQU   31
                                                 2419
                                                 2420           END
```

```
     SYMBOL        TYPE   VALUE      LENGTH    DEFN  REFERENCES

BEGIN               I    00000200         2    162   128   158   159   160
CTLR0               F    0000053C         4    391   172   173   174   175
DECNUM              C    00001073        16    443   305   307   313   315
E7TEST              4    00000000        64    457   254
E7TESTS             F    00003168         4   2297   247
EDIT                X    00001047        18    438   306   314
ENDTEST             U    000003CE         1    291   252
EOJ                 I    00000520         4    381   207   240   294
EOJPSW              D    00000510         8    379   381
FAILCONT            U    000003BE         1    281
FAILED              F    00001000         4    420   283   292
FAILMSG             U    000003BA         1    275   265
FAILPSW             D    00000528         8    383   385
FAILTEST            I    00000538         4    385   295
FB0001              F    00000280         8    191   195   196   198
FB0002              F    00000330         8    224   228   229   231
IMAGE               1    00000000     12884      0
K                   U    00000400         1    404   405   406   407
K64                 U    00010000         1    406
M4                  U    00000007         1    461   312
MB                  U    00100000         1    407
MSG                 I    00000458         4    341   206   239   324
MSGCMD              C    000004A6         9    371   354   355
MSGMSG              C    000004AF        95    372   348   369   346
MSGMVC              I    000004A0         6    369   352
MSGOK               I    0000046E         2    350   347
MSGRET              I    0000048E         4    365   358   361
MSGSAVE             F    00000494         4    368   344   365
NEXTE6              U    00000384         1    249   268   286
OPNAME              C    00000008         8    463   310
PAGE                U    00001000         1    405
PRT3                C    0000105D        18    441   306   307   308   314   315   316
PRTLINE             C    00001008        16    426   433   323
PRTLNG              U    0000003F         1    433   322
PRTM4               C    00001044         2    431   316
PRTNAME             C    00001033         8    429   310
PRTNUM              C    00001018         3    427   308
R0                  U    00000000         1   2366   122   172   175   195   197   198   199   204   228   230   231   232   237
                                                     256   257   282   283   321   322   325   341   344   346   348   350   365
R1                  U    00000001         1   2367   205   238   263   264   292   293   323   355   369   590   591   592   593
                                                     621   622   623   624   652   653   654   655   683   684   685   686   714
                                                     715   716   717   745   746   747   748   776   777   778   779   810   811
                                                     812   813   841   842   843   844   872   873   874   875   903   904   905
                                                     906   934   935   936   937   965   966   967   968   996   997   998   999
                                                    1030  1031  1032  1033  1061  1062  1063  1064  1092  1093  1094  1095  1123
                                                    1124  1125  1126  1155  1156  1157  1158  1186  1187  1188  1189  1217  1218
                                                    1219  1220  1251  1252  1253  1254  1282  1283  1284  1285  1313  1314  1315
                                                    1316  1344  1345  1346  1347  1375  1376  1377  1378  1406  1407  1408  1409
                                                    1437  1438  1439  1440  1471  1472  1473  1474  1502  1503  1504  1505  1533
                                                    1534  1535  1536  1564  1565  1566  1567  1595  1596  1597  1598  1626  1627
                                                    1628  1629  1657  1658  1659  1660  1688  1689  1690  1691  1719  1720  1721
                                                    1722  1749  1750  1751  1752  1779  1780  1781  1782  1809  1810  1811  1812
                                                    1839  1840  1841  1842  1869  1870  1871  1872  1903  1904  1905  1906  1934
                                                    1935  1936  1937  1965  1966  1967  1968  1996  1997  1998  1999  2027  2028
                                                    2029  2030  2058  2059  2060  2061  2089  2090  2091  2092  2120  2121  2122
                                                    2123  2153  2154  2155  2156  2184  2185  2186  2187  2215  2216  2217  2218
```

```
     SYMBOL        TYPE   VALUE      LENGTH    DEFN  REFERENCES

                                                     2246  2247  2248  2249  2277  2278  2279  2280
R10                 U    0000000A           1  2376   160   169   170
R11                 U    0000000B           1  2377   260   261   596   627   658   689   720   751   782   816   847   878   909
                                                      940   971  1002  1036  1067  1098  1129  1161  1192  1223  1257  1288  1319
                                                     1350  1381  1412  1443  1477  1508  1539  1570  1601  1632  1663  1694  1725
                                                     1755  1785  1815  1845  1875  1909  1940  1971  2002  2033  2064  2095  2126
                                                     2159  2190  2221  2252  2283
R12                 U    0000000C           1  2378   247   250   267   285
R13                 U    0000000D           1  2379
R14                 U    0000000E           1  2380
R15                 U    0000000F           1  2381   276   301   328   329
R2                  U    00000002           1  2368   206   239   304   305   312   313   321   324   325   342   344   350   351
                                                      352   354   360   365   366
R3                  U    00000003           1  2369
R4                  U    00000004           1  2370
R5                  U    00000005           1  2371   250   251   254   302   327   575   598   606   629   637   660   668   691
                                                      699   722   730   753   761   784   795   818   826   849   857   880   888
                                                      911   919   942   950   973   981  1004  1015  1038  1046  1069  1077  1100
                                                     1108  1131  1140  1163  1171  1194  1202  1225  1236  1259  1267  1290  1298
                                                     1321  1329  1352  1360  1383  1391  1414  1422  1445  1456  1479  1487  1510
                                                     1518  1541  1549  1572  1580  1603  1611  1634  1642  1665  1673  1696  1704
                                                     1727  1734  1757  1764  1787  1794  1817  1824  1847  1854  1877  1888  1911
                                                     1919  1942  1950  1973  1981  2004  2012  2035  2043  2066  2074  2097  2105
                                                     2128  2138  2161  2169  2192  2200  2223  2231  2254  2262  2285
R6                  U    00000006           1  2372
R7                  U    00000007           1  2373
R8                  U    00000008           1  2374   158   162   163   164   166
R9                  U    00000009           1  2375   159   166   167   169
RE1                 F    00001120           4   597   581   582   584
RE10                F    00001678           4   879   863   864   866
RE11                F    00001710           4   910   894   895   897
RE12                F    000017A8           4   941   925   926   928
RE13                F    00001840           4   972   956   957   959
RE14                F    000018D8           4  1003   987   988   990
RE15                F    00001970           4  1037  1021  1022  1024
RE16                F    00001A08           4  1068  1052  1053  1055
RE17                F    00001AA0           4  1099  1083  1084  1086
RE18                F    00001B38           4  1130  1114  1115  1117
RE19                F    00001BD0           4  1162  1146  1147  1149
RE2                 F    000011B8           4   628   612   613   615
RE20                F    00001C68           4  1193  1177  1178  1180
RE21                F    00001D00           4  1224  1208  1209  1211
RE22                F    00001D98           4  1258  1242  1243  1245
RE23                F    00001E30           4  1289  1273  1274  1276
RE24                F    00001EC8           4  1320  1304  1305  1307
RE25                F    00001F60           4  1351  1335  1336  1338
RE26                F    00001FF8           4  1382  1366  1367  1369
RE27                F    00002090           4  1413  1397  1398  1400
RE28                F    00002128           4  1444  1428  1429  1431
RE29                F    000021C0           4  1478  1462  1463  1465
RE3                 F    00001250           4   659   643   644   646
RE30                F    00002258           4  1509  1493  1494  1496
RE31                F    000022F0           4  1540  1524  1525  1527
RE32                F    00002388           4  1571  1555  1556  1558
RE33                F    00002420           4  1602  1586  1587  1589
RE34                F    000024B8           4  1633  1617  1618  1620
```

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES		
RE35	F	00002550	4	1664	1648	1649	1651
RE36	F	000025E8	4	1695	1679	1680	1682
RE37	F	00002680	4	1726	1710	1711	1713
RE38	F	00002718	4	1756	1740	1741	1743
RE39	F	000027B0	4	1786	1770	1771	1773
RE4	F	000012E8	4	690	674	675	677
RE40	F	00002848	4	1816	1800	1801	1803
RE41	F	000028E0	4	1846	1830	1831	1833
RE42	F	00002978	4	1876	1860	1861	1863
RE43	F	00002A10	4	1910	1894	1895	1897
RE44	F	00002AA8	4	1941	1925	1926	1928
RE45	F	00002B40	4	1972	1956	1957	1959
RE46	F	00002BD8	4	2003	1987	1988	1990
RE47	F	00002C70	4	2034	2018	2019	2021
RE48	F	00002D08	4	2065	2049	2050	2052
RE49	F	00002DA0	4	2096	2080	2081	2083
RE5	F	00001380	4	721	705	706	708
RE50	F	00002E38	4	2127	2111	2112	2114
RE51	F	00002ED0	4	2160	2144	2145	2147
RE52	F	00002F68	4	2191	2175	2176	2178
RE53	F	00003000	4	2222	2206	2207	2209
RE54	F	00003098	4	2253	2237	2238	2240
RE55	F	00003130	4	2284	2268	2269	2271
RE6	F	00001418	4	752	736	737	739
RE7	F	000014B0	4	783	767	768	770
RE8	F	00001548	4	817	801	802	804
RE9	F	000015E0	4	848	832	833	835
REA1	A	000010D4	4	584			
REA10	A	0000162C	4	866			
REA11	A	000016C4	4	897			
REA12	A	0000175C	4	928			
REA13	A	000017F4	4	959			
REA14	A	0000188C	4	990			
REA15	A	00001924	4	1024			
REA16	A	000019BC	4	1055			
REA17	A	00001A54	4	1086			
REA18	A	00001AEC	4	1117			
REA19	A	00001B84	4	1149			
REA2	A	0000116C	4	615			
REA20	A	00001C1C	4	1180			
REA21	A	00001CB4	4	1211			
REA22	A	00001D4C	4	1245			
REA23	A	00001DE4	4	1276			
REA24	A	00001E7C	4	1307			
REA25	A	00001F14	4	1338			
REA26	A	00001FAC	4	1369			
REA27	A	00002044	4	1400			
REA28	A	000020DC	4	1431			
REA29	A	00002174	4	1465			
REA3	A	00001204	4	646			
REA30	A	0000220C	4	1496			
REA31	A	000022A4	4	1527			
REA32	A	0000233C	4	1558			
REA33	A	000023D4	4	1589			
REA34	A	0000246C	4	1620			
REA35	A	00002504	4	1651			

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES	
REA36	A	0000259C	4	1682		
REA37	A	00002634	4	1713		
REA38	A	000026CC	4	1743		
REA39	A	00002764	4	1773		
REA4	A	0000129C	4	677		
REA40	A	000027FC	4	1803		
REA41	A	00002894	4	1833		
REA42	A	0000292C	4	1863		
REA43	A	000029C4	4	1897		
REA44	A	00002A5C	4	1928		
REA45	A	00002AF4	4	1959		
REA46	A	00002B8C	4	1990		
REA47	A	00002C24	4	2021		
REA48	A	00002CBC	4	2052		
REA49	A	00002D54	4	2083		
REA5	A	00001334	4	708		
REA50	A	00002DEC	4	2114		
REA51	A	00002E84	4	2147		
REA52	A	00002F1C	4	2178		
REA53	A	00002FB4	4	2209		
REA54	A	0000304C	4	2240		
REA55	A	000030E4	4	2271		
REA6	A	000013CC	4	739		
REA7	A	00001464	4	770		
REA8	A	000014FC	4	804		
REA9	A	00001594	4	835		
READDR	A	0000001C	4	467	263	
REG2LOW	U	000000DD	1	410		
REG2PATT	U	AABBCCDD	1	409		
RELEN	A	00000018	4	466		
RPTDWSAV	D	00000448	8	334	321	325
RPTERROR	I	000003DC	4	301	276	
RPTSAVE	F	00000440	4	331	301	328
RPTSVR5	F	00000444	4	332	302	327
SKL0001	U	0000004E	1	188	204	
SKL0002	U	0000004E	1	221	237	
SKT0001	C	0000022A	20	185	188	205
SKT0002	C	000002D4	20	218	221	238
SVOLDPSW	U	00000140	0	124		
T1	A	000010B8	4	576	2300	
T10	A	00001610	4	858	2309	
T11	A	000016A8	4	889	2310	
T12	A	00001740	4	920	2311	
T13	A	000017D8	4	951	2312	
T14	A	00001870	4	982	2313	
T15	A	00001908	4	1016	2314	
T16	A	000019A0	4	1047	2315	
T17	A	00001A38	4	1078	2316	
T18	A	00001AD0	4	1109	2317	
T19	A	00001B68	4	1141	2318	
T2	A	00001150	4	607	2301	
T20	A	00001C00	4	1172	2319	
T21	A	00001C98	4	1203	2320	
T22	A	00001D30	4	1237	2321	
T23	A	00001DC8	4	1268	2322	
T24	A	00001E60	4	1299	2323	

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES	
T25	A	00001EF8	4	1330	2324	
T26	A	00001F90	4	1361	2325	
T27	A	00002028	4	1392	2326	
T28	A	000020C0	4	1423	2327	
T29	A	00002158	4	1457	2328	
T3	A	000011E8	4	638	2302	
T30	A	000021F0	4	1488	2329	
T31	A	00002288	4	1519	2330	
T32	A	00002320	4	1550	2331	
T33	A	000023B8	4	1581	2332	
T34	A	00002450	4	1612	2333	
T35	A	000024E8	4	1643	2334	
T36	A	00002580	4	1674	2335	
T37	A	00002618	4	1705	2336	
T38	A	000026B0	4	1735	2337	
T39	A	00002748	4	1765	2338	
T4	A	00001280	4	669	2303	
T40	A	000027E0	4	1795	2339	
T41	A	00002878	4	1825	2340	
T42	A	00002910	4	1855	2341	
T43	A	000029A8	4	1889	2342	
T44	A	00002A40	4	1920	2343	
T45	A	00002AD8	4	1951	2344	
T46	A	00002B70	4	1982	2345	
T47	A	00002C08	4	2013	2346	
T48	A	00002CA0	4	2044	2347	
T49	A	00002D38	4	2075	2348	
T5	A	00001318	4	700	2304	
T50	A	00002DD0	4	2106	2349	
T51	A	00002E68	4	2139	2350	
T52	A	00002F00	4	2170	2351	
T53	A	00002F98	4	2201	2352	
T54	A	00003030	4	2232	2353	
T55	A	000030C8	4	2263	2354	
T6	A	000013B0	4	731	2305	
T7	A	00001448	4	762	2306	
T8	A	000014E0	4	796	2307	
T9	A	00001578	4	827	2308	
TESTING	F	00001004	4	421	257	
TNUM	H	00000004	2	459	256	304
TSUB	A	00000000	4	458	260	
TTABLE	F	00003168	4	2299		
V0	U	00000000	1	2387		
V1	U	00000001	1	2388	259	
V10	U	0000000A	1	2397		
V11	U	0000000B	1	2398		
V12	U	0000000C	1	2399		
V13	U	0000000D	1	2400		
V14	U	0000000E	1	2401		
V15	U	0000000F	1	2402		
V16	U	00000010	1	2403		
V17	U	00000011	1	2404		
V18	U	00000012	1	2405		
V19	U	00000013	1	2406		
V1FUDGE	X	00001094	16	450	259	
V1O1	X	000010E0	16	586	595	

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
V1010	X	00001638	16	868	877
V1011	X	000016D0	16	899	908
V1012	X	00001768	16	930	939
V1013	X	00001800	16	961	970
V1014	X	00001898	16	992	1001
V1015	X	00001930	16	1026	1035
V1016	X	000019C8	16	1057	1066
V1017	X	00001A60	16	1088	1097
V1018	X	00001AF8	16	1119	1128
V1019	X	00001B90	16	1151	1160
V102	X	00001178	16	617	626
V1020	X	00001C28	16	1182	1191
V1021	X	00001CC0	16	1213	1222
V1022	X	00001D58	16	1247	1256
V1023	X	00001DF0	16	1278	1287
V1024	X	00001E88	16	1309	1318
V1025	X	00001F20	16	1340	1349
V1026	X	00001FB8	16	1371	1380
V1027	X	00002050	16	1402	1411
V1028	X	000020E8	16	1433	1442
V1029	X	00002180	16	1467	1476
V103	X	00001210	16	648	657
V1030	X	00002218	16	1498	1507
V1031	X	000022B0	16	1529	1538
V1032	X	00002348	16	1560	1569
V1033	X	000023E0	16	1591	1600
V1034	X	00002478	16	1622	1631
V1035	X	00002510	16	1653	1662
V1036	X	000025A8	16	1684	1693
V1037	X	00002640	16	1715	1724
V1038	X	000026D8	16	1745	1754
V1039	X	00002770	16	1775	1784
V104	X	000012A8	16	679	688
V1040	X	00002808	16	1805	1814
V1041	X	000028A0	16	1835	1844
V1042	X	00002938	16	1865	1874
V1043	X	000029D0	16	1899	1908
V1044	X	00002A68	16	1930	1939
V1045	X	00002B00	16	1961	1970
V1046	X	00002B98	16	1992	2001
V1047	X	00002C30	16	2023	2032
V1048	X	00002CC8	16	2054	2063
V1049	X	00002D60	16	2085	2094
V105	X	00001340	16	710	719
V1050	X	00002DF8	16	2116	2125
V1051	X	00002E90	16	2149	2158
V1052	X	00002F28	16	2180	2189
V1053	X	00002FC0	16	2211	2220
V1054	X	00003058	16	2242	2251
V1055	X	000030F0	16	2273	2282
V106	X	000013D8	16	741	750
V107	X	00001470	16	772	781
V108	X	00001508	16	806	815
V109	X	000015A0	16	837	846
V1OUTPUT	X	00000028	16	469	264
V2	U	00000002	1	2389	

```
     SYMBOL        TYPE   VALUE      LENGTH    DEFN  REFERENCES

V20                 U    00000014           1  2407
V21                 U    00000015           1  2408
V22                 U    00000016           1  2409   591   594   595   622   625   626   653   656   657   684   687   688   715
                                                      718   719   746   749   750   777   780   781   811   814   815   842   845
                                                      846   873   876   877   904   907   908   935   938   939   966   969   970
                                                      997  1000  1001  1031  1034  1035  1062  1065  1066  1093  1096  1097  1124
                                                     1127  1128  1156  1159  1160  1187  1190  1191  1218  1221  1222  1252  1255
                                                     1256  1283  1286  1287  1314  1317  1318  1345  1348  1349  1376  1379  1380
                                                     1407  1410  1411  1438  1441  1442  1472  1475  1476  1503  1506  1507  1534
                                                     1537  1538  1565  1568  1569  1596  1599  1600  1627  1630  1631  1658  1661
                                                     1662  1689  1692  1693  1720  1723  1724  1750  1753  1754  1780  1783  1784
                                                     1810  1813  1814  1840  1843  1844  1870  1873  1874  1904  1907  1908  1935
                                                     1938  1939  1966  1969  1970  1997  2000  2001  2028  2031  2032  2059  2062
                                                     2063  2090  2093  2094  2121  2124  2125  2154  2157  2158  2185  2188  2189
                                                     2216  2219  2220  2247  2250  2251  2278  2281  2282
V23                 U    00000017           1  2410   593   594   624   625   655   656   686   687   717   718   748   749   779
                                                      780   813   814   844   845   875   876   906   907   937   938   968   969
                                                      999  1000  1033  1034  1064  1065  1095  1096  1126  1127  1158  1159  1189
                                                     1190  1220  1221  1254  1255  1285  1286  1316  1317  1347  1348  1378  1379
                                                     1409  1410  1440  1441  1474  1475  1505  1506  1536  1537  1567  1568  1598
                                                     1599  1629  1630  1660  1661  1691  1692  1722  1723  1752  1753  1782  1783
                                                     1812  1813  1842  1843  1872  1873  1906  1907  1937  1938  1968  1969  1999
                                                     2000  2030  2031  2061  2062  2092  2093  2123  2124  2156  2157  2187  2188
                                                     2218  2219  2249  2250  2280  2281
V24                 U    00000018           1  2411
V25                 U    00000019           1  2412
V26                 U    0000001A           1  2413
V27                 U    0000001B           1  2414
V28                 U    0000001C           1  2415
V29                 U    0000001D           1  2416
V2ADDR              A    00000010           4   464   590   621   652   683   714   745   776   810   841   872   903   934   965
                                                      996  1030  1061  1092  1123  1155  1186  1217  1251  1282  1313  1344  1375
                                                     1406  1437  1471  1502  1533  1564  1595  1626  1657  1688  1719  1749  1779
                                                     1809  1839  1869  1903  1934  1965  1996  2027  2058  2089  2120  2153  2184
                                                     2215  2246  2277
V3                  U    00000003           1  2390
V30                 U    0000001E           1  2417
V31                 U    0000001F           1  2418
V3ADDR              A    00000014           4   465   592   623   654   685   716   747   778   812   843   874   905   936   967
                                                      998  1032  1063  1094  1125  1157  1188  1219  1253  1284  1315  1346  1377
                                                     1408  1439  1473  1504  1535  1566  1597  1628  1659  1690  1721  1751  1781
                                                     1811  1841  1871  1905  1936  1967  1998  2029  2060  2091  2122  2155  2186
                                                     2217  2248  2279
V4                  U    00000004           1  2391
V5                  U    00000005           1  2392
V6                  U    00000006           1  2393
V7                  U    00000007           1  2394
V8                  U    00000008           1  2395
V9                  U    00000009           1  2396
X0001               U    000002A8           1   194   182   195
X0002               U    00000358           1   227   215   228
X1                  F    000010F8           4   589   576
X10                 F    00001650           4   871   858
X11                 F    000016E8           4   902   889
X12                 F    00001780           4   933   920
X13                 F    00001818           4   964   951
```

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
X14	F	000018B0	4	995	982
X15	F	00001948	4	1029	1016
X16	F	000019E0	4	1060	1047
X17	F	00001A78	4	1091	1078
X18	F	00001B10	4	1122	1109
X19	F	00001BA8	4	1154	1141
X2	F	00001190	4	620	607
X20	F	00001C40	4	1185	1172
X21	F	00001CD8	4	1216	1203
X22	F	00001D70	4	1250	1237
X23	F	00001E08	4	1281	1268
X24	F	00001EA0	4	1312	1299
X25	F	00001F38	4	1343	1330
X26	F	00001FD0	4	1374	1361
X27	F	00002068	4	1405	1392
X28	F	00002100	4	1436	1423
X29	F	00002198	4	1470	1457
X3	F	00001228	4	651	638
X30	F	00002230	4	1501	1488
X31	F	000022C8	4	1532	1519
X32	F	00002360	4	1563	1550
X33	F	000023F8	4	1594	1581
X34	F	00002490	4	1625	1612
X35	F	00002528	4	1656	1643
X36	F	000025C0	4	1687	1674
X37	F	00002658	4	1718	1705
X38	F	000026F0	4	1748	1735
X39	F	00002788	4	1778	1765
X4	F	000012C0	4	682	669
X40	F	00002820	4	1808	1795
X41	F	000028B8	4	1838	1825
X42	F	00002950	4	1868	1855
X43	F	000029E8	4	1902	1889
X44	F	00002A80	4	1933	1920
X45	F	00002B18	4	1964	1951
X46	F	00002BB0	4	1995	1982
X47	F	00002C48	4	2026	2013
X48	F	00002CE0	4	2057	2044
X49	F	00002D78	4	2088	2075
X5	F	00001358	4	713	700
X50	F	00002E10	4	2119	2106
X51	F	00002EA8	4	2152	2139
X52	F	00002F40	4	2183	2170
X53	F	00002FD8	4	2214	2201
X54	F	00003070	4	2245	2232
X55	F	00003108	4	2276	2263
X6	F	000013F0	4	744	731
X7	F	00001488	4	775	762
X8	F	00001520	4	809	796
X9	F	000015B8	4	840	827
XC0001	U	000002D0	1	208	200
XC0002	U	00000380	1	241	233
ZVE7TST	J	00000000	12884	121	124 126 130 134 419 122
=A(E7TESTS)	A	0000054C	4	397	247
=AL2(L'MSGMSG)	R	00000556	2	400	346
=F'1'	F	00000550	4	398	282

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
=F'2'	F	00000548	4	396	232
=F'64'	F	00000544	4	395	199
=H'0'	H	00000554	2	399	341

```
 MACRO   DEFN  REFERENCES

FCHECK     74    181    214
PTTABLE   533   2298
VRR_C     488    573    604    635    666    697    728    759    793    824    855    886    917    948    979   1013   1044   1075
                1106   1138   1169   1200   1234   1265   1296   1327   1358   1389   1420   1454   1485   1516   1547   1578   1609
                1640   1671   1702   1732   1762   1792   1822   1852   1886   1917   1948   1979   2010   2041   2072   2103   2136
                2167   2198   2229   2260
```

```
   DESC      SYMBOL   SIZE     POS        ADDR

Entry: 0

Image      IMAGE    12884  0000-3253  0000-3253
  Region            12884  0000-3253  0000-3253
    CSECT  ZVE7TST  12884  0000-3253  0000-3253
```

```
   STMT                           FILE NAME

1     /devstor/sharedvfp/tests/zvector-e7-01-MinMaxAvg.asm

** NO ERRORS FOUND **
```